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# Coarsening Confers Robustness: Causal Bayesian Optimization under DAG Misspecification

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## Abstract

Causal Bayesian optimization (CBO) uses a causal graph to prune the space of candidate interventions and to build informative priors, but requires the entire DAG to be specified correctly, and robustness to structural misspecification is a recognized open problem of the field. We propose coarsening as an input contract: the practitioner supplies only a partition of the manipulable variables and the cluster-level (quotient) graph, and we prove an input-equivalence guarantee — any optimizer whose graph-dependent components factor through this quotient behaves identically under every fine-graph hypothesis with the same quotient, so no misspecification confined within a cluster can affect its exploration sets, priors, surrogates, or trajectory. We instantiate the contract for the three main families of the field: QCBO, which applies CBO’s own minimal-intervention-set rule to the C-DAG, identifiability deciding each arm’s prior tier (a do-calculus prior, or a common uninformative one) rather than its membership; QDCBO, which quotients dynamic CBO slice by slice; and QMCBO, which coarsens model-based CBO’s mechanism factorization into joint cluster mechanisms. Each is invariant, trajectory for trajectory, under within-cluster edge errors that measurably shift its full-graph counterpart, and each reduces exactly to its base method at the identity partition. The guarantee protects behavior, not solution quality: when the optimum splits a cluster, a fixed partition pays regret linear in the horizon, and plateau-triggered refinement down a declared hier-

archy restores convergence to the optimum. The trade is intrinsic: sub-cluster effects are provably not identifiable from quotient-level inputs alone — resolution must be bought with exposure.

## 1 Introduction

Causal Bayesian optimization [Aglietti et al., 2020] unifies two ideas: when the variable  $Y$  we wish to optimize sits inside a causal system, the right search space is the space of interventions  $\text{do}(\mathbf{X}_s = \mathbf{x}_s)$ , and observational data plus the causal graph yield informative priors over the interventional response surfaces. The catch is the input requirement: CBO and its descendants [Aglietti et al., 2021, 2023, Sussex et al., 2023] assume the full DAG is specified correctly, and the survey of Anonymous [2026] ranks robustness to causal-assumption violations first among the field’s open problems, with stress tests showing that method rankings can change substantially when only the assumed graph is perturbed. We study *graph misspecification*: the learner is given, and trusts, a single graph that differs from the truth (as opposed to graph uncertainty, where the learner maintains a posterior). Our thesis is that this fragility is best addressed by committing to less structure of a more reliable kind.

**A wrong edge is costly.** The assumed graph enters a CBO algorithm only through its derived causal-effect estimators, and it corrupts them in two ways: it biases the do-calculus prior, and it can change the search set itself, possibly removing the candidate that contains the optimum (Remark 2). Either way the damage is a functional of the wrong edge: an estimator that cannot see an edge cannot be damaged by it.

**Coarser commitments decide less, so less can be wrong.** A cluster-level structural belief encodes strictly fewer decisions than a full DAG: one aggregate judgement between two groups of variables stands in

for a bundle of edge-level judgements the modeler is never asked to make. Both ends of this spectrum are served: full-DAG CBO at one end, methods that learn structure online at heavy per-trial cost at the other [Mukherjee et al., 2024, Branchini et al., 2023]. The practical middle, reliable cluster-level structure, has a developed identification theory [Anand et al., 2023, Ferreira and Assaad, 2025, Wahl et al., 2024] but, to our knowledge, no optimization method.

**Quotient CBO.** QCBO fills this gap. The practitioner supplies a partition  $\Pi$  of the manipulable variables and the C-DAG  $\mathcal{G}^\Pi$  over its clusters (when a fine-grained hypothesis graph is available,  $\mathcal{G}^\Pi$  is its quotient, with bidirected edges absorbing latent confounding; Lee and Bareinboim, 2019). QCBO explores cluster-level interventions — the minimal intervention sets of the C-DAG, the same rule CBO applies to a full DAG — using identified cluster-level do-calculus estimates as GP prior means where the quotient identifies them, and a common uninformative prior where it does not. This input never looks inside a cluster: an edge error confined to a cluster cannot even be represented, let alone affect the optimizer.

## Contributions.

- **Robustness as an input contract.** Quotient invariance is *method-agnostic* (Theorem 1): any optimizer whose graph-dependent components factor through  $(\Pi, \mathcal{G}^\Pi)$  is invariant to arbitrary within-cluster edge errors. We instantiate the contract for the arm-based, dynamic, and model-based families (QCBO, QDCBO, QMCBO; §4).
- **Exploration focus.** Coarsening concentrates exploration on fewer cluster-level arms; Prop. 3 characterizes its cost.

## 2 Background

A structural causal model  $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, F, P(\mathbf{U}) \rangle$  induces a distribution over observables  $\mathbf{V}$  and, for any  $\mathbf{X}_s \subseteq \mathbf{V}$ , interventional distributions  $P(\cdot \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s))$  obtained by replacing the mechanisms of  $\mathbf{X}_s$  with constants [Pearl, 2009]. The induced graph  $\mathcal{G}$  has a directed edge per functional argument and a bidirected edge per shared latent; projecting out latents yields an ADMG over  $\mathbf{V}$  [Lee and Bareinboim, 2019].

**Causal Bayesian optimization.** Given manipulable variables  $\mathbf{X} \subseteq \mathbf{V}$  and target  $Y$ , CBO [Aglietti et al., 2020] solves

$$(\mathbf{X}_s^*, \mathbf{x}_s^*) = \arg \min_{\substack{\mathbf{X}_s \in \mathbf{ES} \\ \mathbf{x}_s \in D(\mathbf{X}_s)}} \mathbb{E}[Y \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s)], \quad (1)$$

where  $D(\mathbf{X}_s)$  is the interventional domain, maintaining one GP per candidate set  $\mathbf{X}_s$  in an exploration set  $\mathbf{ES}$  (each a continuous BO sub-problem over  $\mathbf{x}_s$ ), with prior mean the do-calculus estimate from observational data and the assumed graph. The exploration set is pruned structurally: CBO searches over the *minimal intervention sets* (MIS) of the assumed graph [Lee and Bareinboim, 2018, Aglietti et al., 2020]. Writing  $\mathcal{G}_{\bar{\mathbf{X}}_s}$  for the mutilated graph with all edges into  $\mathbf{X}_s$  removed and  $\text{An}(\cdot)$  for ancestors,

$$\mathbf{ES} = \text{MIS}(\mathcal{G}) = \{ \emptyset \neq \mathbf{X}_s \subseteq \mathbf{X} : \mathbf{X}_s \subseteq \text{An}(Y) \text{ in } \mathcal{G}_{\bar{\mathbf{X}}_s} \} : \quad (2)$$

if some member of  $\mathbf{X}_s$  is no ancestor of  $Y$  once the whole set is intervened on,  $\text{do}(\mathbf{X}_s)$  induces the same distribution as intervening on a strict subset, so the arm is redundant. The possibly-optimal sets POMIS are a further graph-only pruning with  $\text{POMIS} \subseteq \text{MIS}$  [Lee and Bareinboim, 2018]. Each trial selects the arm and value maximizing cost-normalized causal expected improvement: with incumbent  $y^{\text{best}}$  and per-arm posterior  $(\mu_s, \sigma_s)$ ,

$$\text{EI}_s(\mathbf{x}) = \sigma_s(\mathbf{x}) [\gamma_s(\mathbf{x}) \Phi(\gamma_s(\mathbf{x})) + \phi(\gamma_s(\mathbf{x}))], \quad \gamma_s(\mathbf{x}) = \frac{y^{\text{best}} - \mu_s(\mathbf{x})}{\sigma_s(\mathbf{x})}, \quad (3)$$

the closed form of  $\mathbb{E}[\max(y^{\text{best}} - Y_s(\mathbf{x}), 0)]$  under the GP posterior [Garnett, 2023], divided by the intervention cost  $\text{Co}(\mathbf{X}_s, \mathbf{x})$  at selection. The model-based family selects instead by GP-UCB,

$$\alpha_s(\mathbf{x}) = \mu_s(\mathbf{x}) + \sqrt{\beta} \sigma_s(\mathbf{x}), \quad (4)$$

with exploration weight  $\beta$  ( $\beta = 10$  in all our runs, MCBO’s default; Sussex et al., 2023). Both the pruning and the priors are functions of the assumed graph, which is where misspecification enters.

## Coarsening and C-DAGs.

**Definition 1** (Coarsening [Madaleno et al., 2026a]). *Given a DAG  $\mathcal{G}$  on  $\mathbf{V}$ , a partition  $\Pi = \{C_1, \dots, C_K\}$  of  $\mathbf{V}$  induces the surjection  $\chi : \mathbf{V} \rightarrow \Pi$  mapping each vertex to its cluster, and the quotient graph  $\mathcal{G}^\Pi$  with vertex set  $\Pi$  and edge set  $\{\chi(V_i) \rightarrow \chi(V_j) \mid V_i \rightarrow V_j \in \mathcal{G}, \chi(V_i) \neq \chi(V_j)\}$ . If  $\mathcal{G}^\Pi$  is acyclic, it is a **coarsening** of  $\mathcal{G}$  and  $\Pi$  is valid.*

For ADMGs the quotient additionally carries  $C_k \leftrightarrow C_\ell$  iff some  $V_i \in C_k, V_j \in C_\ell$  share a bidirected edge. The result is a C-DAG in the sense of Anand et al. [2023]: do-calculus on it is sound and complete for the class of fine graphs that quotient to it. Markov properties transfer to the cluster level, faithfulness does not [Wahl et al., 2024].

**Evaluation metrics.** Primary metrics are direct functions of the trajectory and the oracle optimum

$y^*$ : the best value found  $y_T^{\text{best}}$  (final  $Y$ ), the simple regret of the incumbent  $r_t = |y_t^{\text{best}} - y^*|$ , and the cumulative regret  $R_T = \sum_{t=1}^T r_t$ . As a secondary diagnostic we report GAP, introduced with DCBO [Aglietti et al., 2021]: with  $y_0$  the initial incumbent,  $\rho_t = \min\{(y_0 - y_t^{\text{best}})/(y_0 - y^*), 1\}$  the clipped improvement ratio (minimization shown), and  $t^*$  the first trial attaining the final ratio  $\rho_T$ ,

$$\text{GAP} = \frac{\rho_T + (T - t^*)/T}{1 + (T - 1)/T}.$$

GAP combines the final improvement with how early it was attained (higher is better;  $\text{GAP}@k$  is GAP on the first  $k$  trials), but it depends on the method-specific initial incumbent  $y_0$ , so final  $Y$  and regret carry the headline comparisons.

### 3 What coarsening buys

Coarsening is not a property of one algorithm, so we state it as an input contract. An optimizer accepts the pair  $(\Pi, \mathcal{G}^\Pi)$  in place of a fine graph, and every graph-dependent component (search space, identification formulas, priors, surrogate architecture, selection rules) is required to be a function of that pair. Any optimizer meeting the contract inherits the invariance below, whatever its family. Read the theorem for what it is: an *input-equivalence* statement. Two hypotheses with the same quotient are indistinguishable to the optimizer, so no intra-cluster mistake can change its behavior; the theorem does not certify solution quality, which rests on the quotient itself being correct. A wrong quotient produces the same trajectory for the wrong problem (§5.4 measures that cost).

Throughout,  $\Pi, \Pi'$  are manipulable-only partitions.  $\Pi_\perp$  is the finest (all-singleton) partition, the bottom of the partition refinement lattice [Madaleno et al., 2026a], and  $\Pi' \preceq \Pi$  ( $\Pi'$  refines  $\Pi$ ) iff every cluster of  $\Pi'$  is contained in a cluster of  $\Pi$ . Props. 1–3 relate quotients of a common underlying graph. The instantiations of §4 consume only  $\mathcal{G}^\Pi$  (Assumption 1).

**Theorem 1** (Quotient invariance is method-agnostic). *Let  $\mathcal{A}$  be any causal Bayesian optimizer whose graph-dependent components (search space, identification formulas, priors, surrogate architecture, and selection rules) depend on the structural hypothesis  $\mathcal{G}$  only through the pair  $(\Pi, \mathcal{G}^\Pi)$ , where  $\mathcal{G}^\Pi$  is the quotient of  $\mathcal{G}$ ’s latent projection under a valid  $\Pi$ . Then for any two acyclic hypotheses  $\mathcal{G}, \mathcal{G}'$  with  $(\mathcal{G}')^\Pi = \mathcal{G}^\Pi$ , and the same data, seeds, and hyperparameters,  $\mathcal{A}$  produces identical trajectories under  $\mathcal{G}$  and  $\mathcal{G}'$ . In particular, this holds for every edit (insertion, deletion, or reversal of directed or bidirected edges) confined within a single cluster, and for inter-cluster edits that are redundant in the quotient. (Proof: App. A)*

**Remark 1** (Exact scope). *A quotient-level input cannot represent an intra-cluster hypothesis, so no intra-cluster mistake can affect the optimizer. The precise condition is that the perturbed edge be invisible in the quotient: intra-cluster edges, and inter-cluster edges made redundant by a parallel edge. A quotient-visible edge is not protected (§5.3).*

**Remark 2** (How a wrong graph enters the algorithm). *A wrong assumed graph reaches any causal Bayesian optimizer through its derived causal-effect estimators, in two ways. Prior bias: the do-calculus prior mean of any arm whose estimand references the edge is biased even though the true effect is unchanged, a convergence cost paid in regret; in QCBO a wrong graph can also flip an arm’s identifiability verdict and drop it to the uninformative prior tier (Eq. 6). Search-set change: the exploration set is itself derived from the graph, by MIS pruning (Eq. 2) in CBO [Lee and Bareinboim, 2018, Aglietti et al., 2020] and its C-DAG analogue (Eq. 5) in QCBO, so a wrong graph can remove an arm, possibly the optimal one, or admit a spurious one. Coarsening closes both channels whenever the edges are quotient-invisible (Theorem 1): the quotient never references the edge, so no estimator it forms is biased and no verdict it makes is flipped.*

**Proposition 1** (Validity is not refinement-monotone). *There exist a valid  $\Pi$  and a refinement  $\Pi' \preceq \Pi$  whose quotient is cyclic; validity must be re-verified after each refinement (it remains decidable from cluster-level structure alone).*

**Proposition 2** (Refinement: values are monotone, membership is not). *Let  $\Pi' \preceq \Pi$  with both valid. (i) Every  $\mathbf{X}_s \in \mathbf{ES}^\Pi$  that is again a MIS of  $\mathcal{G}^{\Pi'}$  lies in  $\mathbf{ES}^{\Pi'}$  with the same identification functional. (ii) Membership itself is not refinement-monotone: a coarse cluster-union arm need not be minimal at the finer partition. (On ToyGraph,  $X \cup Z$  is the single MIS of the coarse quotient, while  $\{X, Z\}$  is not a MIS at  $\Pi_\perp$ : mutilating  $Z$  cuts  $X \rightarrow Z$ , leaving  $X$  causally inert, so  $\text{do}(X, Z) \equiv \text{do}(Z)$ .) What refinement does preserve is value — Prop. 3 below — because every cluster-union arm’s value is attained by its MIS reduction (Lemma 1, App. A).*

**Proposition 3** (Price of coarsening). *Let  $\mu^*(\mathbf{X}_s) = \min_{\mathbf{x}_s \in D(\mathbf{X}_s)} \mathbb{E}[Y \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s)]$  and  $V(\Pi) = \min_{\mathbf{X}_s \in \mathbf{ES}^\Pi} \mu^*(\mathbf{X}_s)$  — the best value reachable from the exploration set QCBO actually searches (Eq. 5). For any valid  $\Pi' \preceq \Pi$ ,  $V(\Pi_\perp) \leq V(\Pi') \leq V(\Pi)$ , and  $V(\Pi) = V(\Pi_\perp)$  iff some optimizer  $\mathbf{X}_s^*$  of Eq. (1) is a union of  $\Pi$ -clusters. (Proof via Lemma 1: the MIS-restricted minimum equals the minimum over all cluster unions, which is refinement-monotone.)*

**Proposition 4** (When coarsening is free). *Assume (i) full-support intervention domains and (ii) cluster-*

contained confounding: in  $\mathcal{G}^*$ , every backdoor path from a cluster member to  $Y$  passes through another member of the same cluster. Then  $V(\Pi) = V(\Pi_\perp)$  for every valid  $\Pi$ : setting the remaining members to the values they would naturally have taken changes nothing (consistency), so any sub-cluster arm’s value is a mixture over whole-cluster interventions, whose minimum the cluster arm attains. A union-of-clusters optimizer of Eq. (1) then always exists and the price of Prop. 3 is zero. Both premises are necessary (App. A: a policy-box example and a member–target confounding example, each with strictly positive price).

**Corollary 1** (The price is a finite-sample effect). *Under the premises of Prop. 4, if the exploration set contains a cluster arm covering an optimal fine arm and the arm-level optimizer is consistent with every arm sampled infinitely often, QCBO’s incumbent converges to  $y^*$  — whichever prior tier the arm runs on, since the uninformative tier only slows learning, never removes the arm. Coarsening then changes where finite samples concentrate, never the attainable value.*

### 3.1 Refining the partition: Hierarchical QCBO (HQCBO)

When no optimizer of Eq. (1) is a union of  $\Pi$ -clusters, a fixed-partition run pays  $V(\Pi) - y^* > 0$  forever:  $R_T/T \rightarrow V(\Pi) - y^*$ . Prop. 4 delimits when this can happen at all: a positive price requires a constrained intervention domain (the policy box of CLUSTERBENCH10L) or confounding that escapes the cluster. Note the kind of failure this is. With a correct graph, a full-DAG method always retains an optimal arm, and a biased prior is eventually overridden by interventional data. A fixed coarse partition lacks the sub-cluster arm even when its quotient is correct: the gap is a property of the action space, and no amount of data restores it. (Under a wrong graph a full-DAG method can suffer the same uncorrectable arm loss through a flipped verdict, Remark 2; refinement addresses the action-space kind.) Refinement converts this failure back into a correctable one by treating  $\Pi$  as a starting resolution.

The practitioner may declare a *hierarchy*  $H$ : per cluster, an optional sub-partition (default: singletons); no new graph is supplied. HQCBO runs Algorithm 1 at  $\Pi$  and, when the incumbent improves by less than  $\delta$  over  $k$  trials ( $k=5$ ,  $\delta=10^{-3}$ , tuned once and fixed), splits the incumbent arm’s clusters via  $H$ , refusing any split that fails the validity re-check (Prop. 1). On acceptance the exploration set is rebuilt and unioned with every explored arm, data carry over keyed by arm (Prop. 2), and only new arms receive the standard  $n_0$  initialization points (App. H). We evaluate a single refinement step.

**Remark 3** (Phase-scoped invariance). *Any misspecification that is quotient-invisible at every visited partition, in particular inside a cluster that is never split, leaves the entire HQCBO trajectory unchanged, including the trigger time and the refinement decision (per phase this is Theorem 1, and the trigger is a function of the trajectory and  $H$  alone). Refining a cluster spends its invariance: edits inside a split cluster are exposed after the trigger.*

**Remark 4** (Refinement makes the price correctable). *The fixed-partition failure is worth stating mechanically: QCBO only ever executes whole-cluster interventions, so when every optimizer of Eq. (1) sets a strict subset of some cluster, every arm the fixed partition can play has value at least  $V(\Pi) > y^*$ . The per-trial regret is then bounded below by  $V(\Pi) - y^* > 0$  at every  $t$  — no amount of data lowers it, because the deficit lives in the action space, not in the estimates — hence  $R_T = \Theta(T)$ . Refinement changes the action space itself: once the exploration set at some visited partition contains an optimal arm and the base optimizer is consistent on a finite arm set, the incumbent converges to  $y^*$  and the average regret  $R_T/T \rightarrow 0$ . We claim no convergence rate and report the post-trigger behavior empirically (§5.4). The trigger never observes  $y^*$ , so it also fires on lossless partitions; there refinement costs only the new arms’ initialization and, empirically, a small refinement price.*

**Remark 5** (Splitting as coarse-to-fine MIS search). *MIS/POMIS pruning discards arms from the graph alone [Lee and Bareinboim, 2018]. HQCBO descends the partition lattice guided by observed value, stopping at the coarsest visited partition whose cluster-level MIS contains a value-optimal arm: a value-driven search for a partition exposing a minimal optimal intervention set, restricted to the declared hierarchy and the incumbent’s clusters.*

## 4 Instantiations: QCBO, QDCBO, and QMCBO

The contract is met by replacing each graph-coupled component of a base optimizer with its quotient counterpart. We instantiate it for the three main families, in increasing order of graph coupling: CBO (the graph prunes arms and builds priors), DCBO (it additionally carries information across time slices), and MCBO (it is the surrogate’s architecture). Figure 1 shows the structural input each family consumes and its quotient.

### 4.1 Arm-based: quotient CBO (QCBO)

QCBO uses only cluster-level structure: a partition  $\Pi$  of the manipulable variables  $\mathbf{X}$  and a C-DAG  $\mathcal{G}^\Pi$

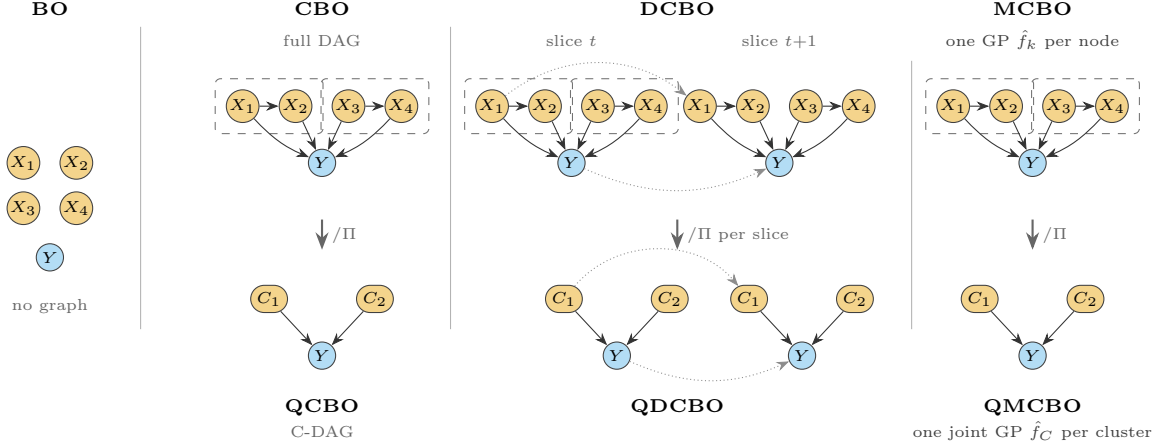


Figure 1: The input contract across the three families. Each base method (CBO, DCBO, MCBO) consumes a full graph; its quotient counterpart consumes only the partition  $\Pi$  and the C-DAG  $\mathcal{G}^\Pi$  ( $/\Pi$  arrows). Non-causal BO (left) consumes no graph. Intra-cluster edges — everything inside a dashed box — do not exist at the quotient level, which is the entire content of Theorem 1.

over its clusters and the singleton nodes.  $Y$  and every non-manipulable observable stay singletons, and mixing manipulable and non-manipulable variables in one cluster is disallowed, since intervening on a cluster (jointly setting all its members,  $\text{do}(C_k = \mathbf{c}_k)$ ) must be executable. When the practitioner instead holds a fine-grained hypothesis graph  $\mathcal{G}$ ,  $\mathcal{G}^\Pi$  is its quotient after projecting out hidden confounders [Lee and Bareinboim, 2019]. QCBO itself never requires the fine graph. The exploration set is CBO’s own rule (Eq. 2) applied to the quotient: the MIS of  $\mathcal{G}^\Pi$  over the manipulable cluster vertices  $\Pi_M$ ,

$$\mathbf{ES}^\Pi = \text{MIS}(\mathcal{G}^\Pi) = \left\{ \bigcup_{C_k \in \mathcal{S}} C_k \mid \emptyset \neq \mathcal{S} \subseteq \Pi_M, \mathcal{S} \subseteq \text{An}(Y) \right\} \quad (5)$$

with  $\text{POMIS}(\mathcal{G}^\Pi) \subseteq \text{MIS}(\mathcal{G}^\Pi)$ , so the possibly-optimal arms are always retained. Merging manipulable variables coarsens the action space: QCBO can only set a cluster jointly, never a strict subset of it. This is the mechanism behind both the robustness (§3) and the price (Prop. 3) of coarsening. Algorithm 1 (App. E) summarizes the loop.

**Identifiability and the two-tier prior.** For each arm  $\mathbf{X}_s \in \mathbf{ES}^\Pi$  we pose the cluster-level query  $P(Y \mid \text{do}(\mathbf{X}_s))$  on  $\mathcal{G}^\Pi$  and run the complete ID algorithm of Shpitser and Pearl [2006] (via `ananke`; Ananke). Identifiability decides the arm’s *prior tier*, never its mem-

bership:

$$m_s(\mathbf{x}_s) = \begin{cases} \hat{\mathbb{E}}[Y \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s)] & \text{query id. from } \mathcal{G}^\Pi \\ & \text{and} \\ & \text{pipeline succeeds,} \\ \bar{Y}_{\text{obs}} & \text{otherwise.} \end{cases} \quad (6)$$

An identifiable arm’s prior mean instantiates the identified formula as a pipeline of GP-based plug-ins (backdoor  $\rightarrow$  front-door  $\rightarrow$   $g$ -computation) on the bidirected-edge expansion of  $\mathcal{G}^\Pi$ . A non-identifiable arm — and any identifiable query the sound-but-incomplete pipeline cannot instantiate — shares the common uninformative prior, the observational mean of  $Y$ , and its interventional response is learned from experimental data alone. The identifiability test is exact where it applies: ID is complete on the C-DAG, and identification from  $\mathcal{G}^\Pi$  is sound for every fine graph in its class [Anand et al., 2023]. Because clusters are manipulable-only and interventions assign full clusters, the cluster-level formulas expand exactly into fine-grained regressions on observational data.

**Assumption 1** (Valid C-DAG; estimator boundary). *The directed part of  $\mathcal{G}^\Pi$  is acyclic, which constrains only inter-cluster structure (intra-cluster acyclicity can even be relaxed; Madaleno et al., 2026b), coincides with validity of the coarsening when  $\mathcal{G}^\Pi$  is a quotient, and is checkable on the cluster-level object. The ID test is complete, while the estimator pipeline behind Eq. (6) is sound but not complete: identifiable queries it misses receive the uninformative tier (in §5 this set is empty).*

**Proposition 5** (Identity recovery). *Under  $\Pi = \Pi_\perp$ ,  $\mathcal{G}^\Pi = \mathcal{G}^*$  and Eq. (5) is exactly Eq. (2): the exploration set is the MIS of the full projected graph and*

the priors are CBO’s do-calculus priors, so QCBO reduces to CBO. (On every benchmark graph of §5 each fine MIS arm is identifiable, so the uninformative tier is vacuous at  $\Pi_\perp$ . App. C documents the two places where the CBO authors’ shipped exploration sets deviate from the MIS definition — a non-minimal arm on ToyGraph and an ad-hoc size truncation on CoralGraph — and the corrected sets both methods use here.)

## 4.2 Dynamic: quotient DCBO (QDCBO)

Dynamic CBO (DCBO; Aglietti et al., 2021) optimizes a sequence of interventions on a dynamic Bayesian network: each slice carries its own exploration sets and priors, built from within-slice (emission) and cross-slice (transition) mechanism estimates on the assumed temporal graph, and slice  $t$ ’s optimum feeds slice  $t+1$ ’s priors. The graph enters the per-slice exploration sets, both estimate families, and the propagation structure.

**Construction (QDCBO).** Given the temporal graph over slices  $t = 0, \dots, T-1$  and a partition  $\Pi$  of the per-slice manipulable variables:

1. *Quotient per slice.* One vertex  $C_{k,t}$  per cluster per slice; emission edges quotient within a slice, transition edges across slices; intra-cluster edges vanish. Validity (acyclicity of each slice’s quotient) is rechecked on construction (Prop. 1).
2. *Exploration sets.* Each slice’s candidate interventions are lifted to whole-cluster patterns — the slice-level analogue of Eq. (5).
3. *Mechanisms.* The emission and transition estimates become joint cluster mechanisms, one multi-output GP per cluster arc (the coregionalization construction of QMCBO below), fit on the per-member observational sequences.
4. *Propagation.* Slice  $t$ ’s optimum feeds slice  $t+1$ ’s priors exactly as in DCBO, now at cluster level.

**Assumptions.** DCBO’s own — known temporal structure with stationary slice topology (or declared change points) and causal sufficiency as used by its estimators — plus the contract’s: manipulable-only clusters and per-slice validity of  $\Pi$ . Singleton clusters reduce every component to stock DCBO.

**Corollary 2** (QDCBO inherits the contract). *Every graph-dependent component of QDCBO is a function of  $(\Pi, \mathcal{G}^\Pi)$  applied per slice, so Theorem 1 applies: quotient-invisible edits, in particular any edit confined to a cluster within a slice, leave its entire trajectory unchanged. At  $\Pi = \Pi_\perp$ , QDCBO is exactly DCBO, and its reachable intervention set at each slice equals QCBO’s cluster-level arms, so the price of coarsening (Prop. 3) applies at each slice.*

## 4.3 Model-based: quotient MCBO (QMCBO)

Model-based CBO (MCBO; Sussex et al., 2023) couples the surrogate to the graph more tightly than CBO: it fits one GP per *mechanism*, evaluates interventions by composing these GPs along the DAG, and selects them by GP-UCB (Eq. 4). The graph is the surrogate’s architecture, so a wrong intra-cluster edge corrupts the factorization every prediction flows through.

**Construction (QMCBO).** Coarsen the factorization: one *joint* mechanism per cluster,  $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$ , modeled as a multi-output GP over the cluster’s member coordinates whose task kernel couples them,

$$k((\mathbf{x}, i), (\mathbf{x}', j)) = k_{\text{in}}(\mathbf{x}, \mathbf{x}') B_{ij}, \quad B \succeq 0,$$

(a coregionalization kernel; Bonilla et al., 2007), composed along the C-DAG with joint within-cluster sampling. Intervention targets are lifted to whole-cluster patterns, and singleton clusters reduce to MCBO’s per-node GPs.

**Corollary 3** (QMCBO inherits the contract). *Every graph-dependent component of QMCBO is a function of  $(\Pi, \mathcal{G}^\Pi)$ , so Theorem 1 applies: quotient-invisible edits leave its entire trajectory unchanged. Moreover, (i) at  $\Pi = \Pi_\perp$ , QMCBO is exactly MCBO; and (ii) QMCBO has no arms, but its reachable intervention set (whole-cluster do patterns) equals QCBO’s cluster-level arms, so Prop. 3 applies: on a lossy partition, no amount of data recovers  $y^*$ .*

**Proposition 6** (Soundness of cluster mechanisms). *Assume causal sufficiency and a valid  $\Pi$ . For every union-of-clusters intervention, the cluster-level truncated factorization (the true joint cluster mechanisms composed along the C-DAG with intervened clusters clamped) equals  $P(\mathbf{V} \mid \text{do}(\mathbf{x}_S))$ . (Proof: App. A.) The joint mechanism is essential: factoring a cluster into per-member single-output GPs discards within-cluster output correlation and is exact only when clusters contain no intra-cluster edges.*

**Proposition 7** (Exposure: sub-cluster interventions are not identifiable from cluster mechanisms). *There exist SCMs  $M_1, M_2$ , causally sufficient and valid for the same  $\Pi$ , with identical joint cluster mechanisms and identical interventional distributions for every union-of-clusters intervention, yet  $P^{M_1}(Y \mid \text{do}(\mathbf{x}_A)) \neq P^{M_2}(Y \mid \text{do}(\mathbf{x}_A))$  for a strict subset  $A \subsetneq C$  of some cluster. (Proof: App. A.) Consequently no quotient-respecting method, arm-based, dynamic, or model-based, can evaluate sub-cluster arms from its inputs alone: the exposure trade of §3.1 is a property of the quotient contract itself, not of our refinement procedure. ToyGraph realizes the arm-based face of this*

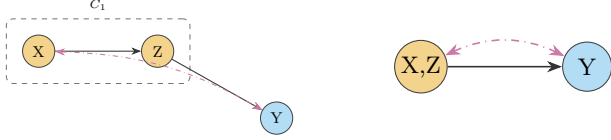


Figure 2: Coarsening can destroy identifiability the fine graph had. ToyGraph’s fine DAG (left):  $X \rightarrow Z \rightarrow Y$  with latent confounding  $X \leftrightarrow Y$ ;  $\text{do}(Z)$  is identifiable (backdoor through  $X$ ) and  $\text{do}(X)$  is identifiable (front-door through  $Z$ ). Its quotient under  $\Pi = \{\{X, Z\}\}$  (right) is a *bow*:  $C_1 \rightarrow Y$  with  $C_1 \leftrightarrow Y$ , so  $P(Y \mid \text{do}(C_1))$  is not identifiable from the C-DAG and the single arm runs on the uninformative prior tier (Eq. 6) — resolution bought with exposure (Prop. 7).

*trade (Fig. 2): merging  $\{X, Z\}$  imports  $X$ ’s latent confounding with  $Y$  into the single cluster arm — the C-DAG is the bow  $C \rightarrow Y$ ,  $C \leftrightarrow Y$  — so  $P(Y \mid \text{do}(C))$  is not identifiable from the quotient even though  $\text{do}(Z)$  is identifiable from the fine graph, and the arm runs on the uninformative prior tier (§5.6).*

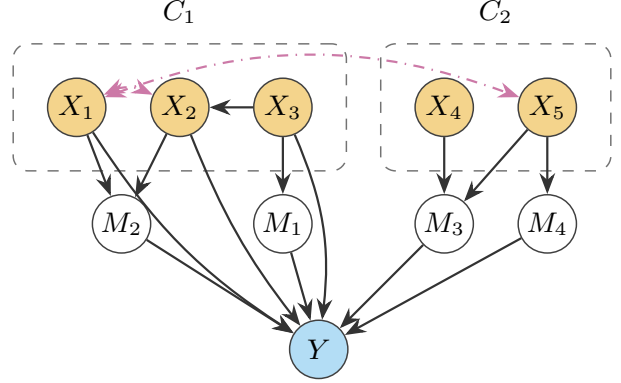
## 5 Experiments

### 5.1 Setup: head-to-head under misspecification

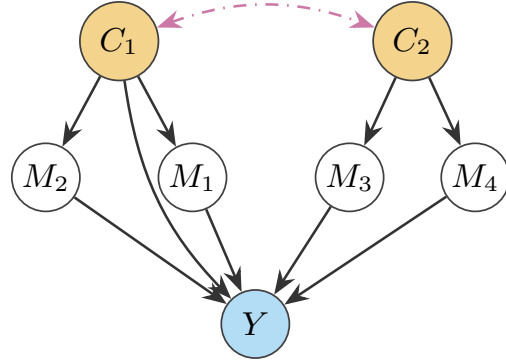
**ClusterBench10.** We construct an SCM (Fig. 3) with ten observables: manipulable  $X_1, \dots, X_5$ , mediators  $M_1, M_2, M_3, M_4$ , target  $Y$ , and latent confounders  $U_{12} \rightarrow \{X_1, X_2\}$  (intra-cluster) and  $U_{15} \rightarrow \{X_1, X_5\}$  (cross-cluster). The coarse partition is  $C_1 = \{X_1, X_2, X_3\}$ ,  $C_2 = \{X_4, X_5\}$ . Coefficients are chosen so that (i)  $\text{do}(X_1)$  is the strongest *singleton* intervention, (ii) the global optimum is  $\text{do}(C_1)$  (a union of clusters, so the partition is *lossless* and any robustness gap is not confounded by a price of coarsening), and (iii)  $\text{corr}(X_1, X_2) \approx 0.70$  (full specification and oracle checks in App. B).

**Fixed-objective protocol.** We hold fixed the true SCM, observational data,  $y^*$ , seeds, and number of trials, and inject the misspecification only into the assumed structure each causal method reasons with, verifying that the injection leaves the objective identical under every perturbation (App. C).

**Perturbations, methods, and metrics.** We perturb the DAG by adding, deleting, or reversing edges, classified as *intra* or *inter* by whether their endpoints share a coarse cluster. The headline perturbation is a *bow*: a directed edge added on a latently-confounded pair, rendering the corresponding do-effect non-identifiable. The arms share one estimator, seeds, and initialization, differing only in the structure each



(a) fine DAG with coarse partition



(b) quotient C-DAG

Figure 3: **ClusterBench10** (orange: manipulable, blue: target, dash-dotted: latent confounding, dashed boxes: clusters). In the quotient (b), the intra-cluster confounding  $X_1 \leftrightarrow X_2$  disappears while the cross-cluster  $X_1 \leftrightarrow X_5$  becomes  $C_1 \leftrightarrow C_2$ . Hence the *intra* bow (adding  $X_1 \rightarrow X_2$ ) is invisible to the quotient, whereas the *inter* bow (adding  $X_1 \rightarrow X_5$ ) completes a cluster-level bow, rendering  $\text{do}(C_1)$  non-identifiable and removing it from the coarse exploration set.

reasons with: BO reads none and optimizes one joint arm under a zero-mean prior, CBO [Aglietti et al., 2020] identifies on the full assumed DAG over its MIS arms (Eq. 2; by Prop. 5 it is QCBO at the finest partition, on every graph), and QCBO runs the same algorithm on the quotient over its cluster-level MIS (Eq. 5), non-identifiable arms on the uninformative prior tier.

We report the final value  $Y_T$ , the simple regret  $r_T = |y_T^{\text{best}} - y^*|$ , and the cumulative regret  $R_T = \sum_{t=1}^T r_t$  of the incumbent sequence as primary metrics, with GAP (§2) as a secondary convergence diagnostic, and score misspecification by the paired-by-seed degradation  $\Delta = \text{metric}(\text{misspecified}, s) - \text{metric}(\text{correct}, s)$ , so per-seed noise cancels.  $\Delta$ -signs are metric-specific:

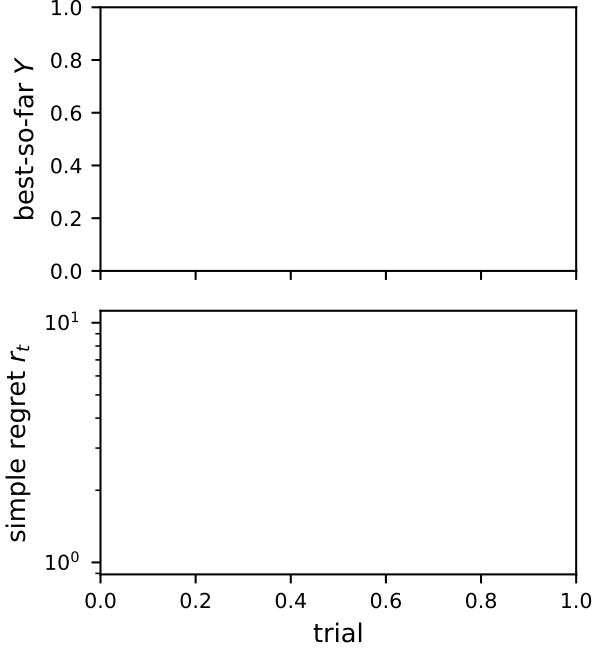


Figure 4: The headline contrast on ClusterBench10: correct DAG (solid) vs intra-cluster bow  $P_1$  (dashed). Bands are  $\pm$ s.e. over 10 seeds. Top: best-so-far  $Y$ . Bottom: simple regret  $r_t$  (log scale). QCBO’s and BO’s dashed curves coincide exactly with their solid ones. The full-DAG method’s dashed curve separates from its solid one, but both converge.

for GAP a positive  $\Delta$  means the misspecified run converged faster, and for  $Y_T$  under minimization a negative  $\Delta$  means it reached a better value.

## 5.2 Coarsening guarantees an unchanged trajectory

We first fix the correct DAG (Table 6): the causal methods exploit the observational prior to converge within a handful of interventions. Non-causal BO, learning the five-dimensional joint intervention response from scratch, is far slower and worst in absolute  $Y$ .

We then add the intra-cluster bow (add  $X_1 \rightarrow X_2$  on the  $U_{12}$ -confounded pair) with the objective held fixed. The damage appears in convergence, where Remark 2 predicts it. For QCBO the bow lies inside  $C_1$ : its trajectory and every score are unchanged ( $\Delta = 0$  exactly, Theorem 1). On the full assumed DAG the bow strips  $\text{do}(X_1)$ , and every superset leaving  $X_2$  unintervened, of identification, so CBO’s behavior shifts (Fig. 4, Table 1).

**The same edit with the opposite sign.** On this SCM the arm loss even *helps*:  $\text{do}(X_1)$  is a “decoy”, so losing it concentrates exploration on the larger arms containing the optimum (Fig. 4). CLUSTERBENCH10D (App. B) moves  $Y$ ’s centers so the strongest singleton *is* the global optimum while the coarse partition stays lossless: the identical bow now costs the cheap optimal arm, and full-DAG CBO’s cumulative regret rises  $4.6\times$  (Table 4, App. B), a per-trial cost no data repaired within the horizon. QCBO is invariant on both variants. Losing a decoy accelerates, losing the optimal arm injures, and which case one is in cannot be known without the very structure in doubt; Theorem 1 removes the question for every quotient-invisible error. (These recorded runs realize the arm loss as removal under the earlier gate rule; see Provenance, App. E.)

## 5.3 Which misspecifications are protected?

A single controlled contrast delimits the scope: the *same* operation, adding a directed edge on a latently-confounded pair. The intra bow (add  $X_1 \rightarrow X_2$ ) leaves QCBO unchanged. The inter bow (add  $X_1 \rightarrow X_5$ , across  $C_1, C_2$ ) forms a *cluster-level* bow  $C_1 \rightarrow C_2$  with  $C_1 \leftrightarrow C_2$ , making  $\text{do}(C_1)$ , the arm QCBO relies on, non-identifiable: the guarantee is lost and the trajectory moves off zero (Table 1). The other quotient-visible perturbation, the inter deletion, is unprotected yet unchanged on this SCM (the deleted edge never enters the identification of an arm the optimizer uses), so quotient-invisibility is *sufficient* for invariance, not necessary. The field also contains the most realistic error of all: omitting the latent confounder  $X_1 \leftrightarrow X_2$  ( $P_7$ ). No verdict flips (the channel is pure prior bias), yet the full-DAG trajectory reshuffles in a seed-dependent direction. The quotient discards intra-cluster bidirected edges, so QCBO is again invariant. The severity dial (Fig. 14, App. F) confirms the pattern across the whole field: QCBO stays at zero across the one-, two-, three-edit intra sweep and leaves zero only under the inter bow.

## 5.4 When the partition is lossy or wrong

**The price of coarsening, realized.** CLUSTERBENCH10L shares CLUSTERBENCH10’s structure but strengthens the  $X_3 \rightarrow X_2$  mechanism and restricts interventions to a policy box, engineered so the global optimum  $\text{do}(X_1, X_3)$  *splits*  $C_1$  (oracle-verified price  $V(\Pi) - y^* = 0.95$ ). The loss is the constrained-domain mechanism of Prop. 4: the mediated path  $X_3 \rightarrow X_2$  drives  $X_2$  to values the box forbids the whole-cluster arm to set directly. This is the regime of Prop. 3: CBO reaches  $y^*$  while QCBO plateaus a bounded gap above it (realized price 1.03, Table 7, App. F). The price is

Table 1: Scope of the guarantee: paired  $\Delta\text{GAP}$  and  $\Delta Y_T$  (misspecified – correct; mean $\pm$ s.e., 10 seeds;  $\Delta Y_T < 0$  means the perturbed run ended closer to  $y^*$ ). QCBO is exactly **0** (identical, bold) on every quotient-invisible perturbation and moves only under the inter bow. Full rows in App. F.

| Perturbation                                | Locus        | CBO                |                    | QCBO               |                    |
|---------------------------------------------|--------------|--------------------|--------------------|--------------------|--------------------|
|                                             |              | $\Delta\text{GAP}$ | $\Delta Y_T$       | $\Delta\text{GAP}$ | $\Delta Y_T$       |
| add $X_1 \rightarrow X_2$ (bow)             | intra        | $+0.116 \pm 0.043$ | $-0.190 \pm 0.052$ | <b>0</b>           | <b>0</b>           |
| del $X_3 \rightarrow X_2$                   | intra        | $+0.156 \pm 0.039$ | $-0.157 \pm 0.052$ | <b>0</b>           | <b>0</b>           |
| rev $X_3 \rightarrow X_2$                   | intra        | $+0.156 \pm 0.039$ | $-0.157 \pm 0.052$ | <b>0</b>           | <b>0</b>           |
| omit $X_1 \leftrightarrow X_2$ (confounder) | intra        | $+0.098 \pm 0.038$ | $-0.171 \pm 0.039$ | <b>0</b>           | <b>0</b>           |
| sweep $k=1$                                 | intra        | $+0.116 \pm 0.043$ | $-0.190 \pm 0.052$ | <b>0</b>           | <b>0</b>           |
| sweep $k=2$                                 | intra        | $+0.120 \pm 0.049$ | $-0.182 \pm 0.055$ | <b>0</b>           | <b>0</b>           |
| sweep $k=3$                                 | intra        | $+0.120 \pm 0.049$ | $-0.181 \pm 0.055$ | <b>0</b>           | <b>0</b>           |
| del $X_1 \rightarrow M_2$                   | inter (red.) | $+0.122 \pm 0.052$ | $-0.198 \pm 0.048$ | <b>0</b>           | <b>0</b>           |
| del $X_3 \rightarrow M_1$                   | inter        | $+0.040 \pm 0.032$ | $-0.091 \pm 0.043$ | <b>0</b>           | <b>0</b>           |
| add $X_1 \rightarrow X_5$ (bow)             | inter        | $+0.007 \pm 0.035$ | $+0.070 \pm 0.073$ | $-0.007 \pm 0.084$ | $+0.538 \pm 0.085$ |

paid only where coarsening is lossy, and is exactly 0 everywhere else (§5.6).

**Refining away the plateau.** HQCBO runs the identical coarse configuration with the singleton hierarchy declared for  $C_1$ . The trigger fires at trial  $11.1 \pm 0.8$  (10 seeds), the split passes the validity recheck, and the warm-started refined phase descends immediately: final  $Y = 0.101 \pm 0.001$ , indistinguishable from CBO ( $0.103 \pm 0.001$ ), against the QCBO plateau at  $1.128 \pm 0.016$ . Cumulative regret (Remark 4): fixed-partition QCBO accrues  $R_{100} = 120.3 \pm 3.1$  and keeps paying forever, while HQCBO’s  $32.1 \pm 2.6$  has stopped growing (Fig. 15, App. H). CBO’s  $5.3 \pm 0.7$  remains the best constant: refinement does not out-run a correctly specified full-DAG optimizer, but it removes fixed-partition QCBO’s linear regret while phase 1 keeps QCBO’s invariance (Remark 3). On the lossless control the trigger still fires and costs nothing ( $R_{100}$ :  $41.6 \pm 5.8$  vs  $41.8 \pm 5.2$ ; App. H).

**The cost of a wrong partition.** Soundness requires the true fine graph to quotient to  $\mathcal{G}^\Pi$ . App. G stresses that assumption with a fixed, acyclic but *incorrect* C-DAG and finds graceful degradation, not collapse (final  $Y = 0.49$  vs  $0.13$  correct,  $0.29$  CBO,  $1.98$  BO): a misspecified *clustering* costs bounded solution quality.

### 5.5 The contract in the model-based family

We run QMCBO against MCBO through MCBO’s own harness and protocol ( $T=100$ , 5 seeds) on its two CBO-lineage tasks (TOYGRAPH and PSAGRAPH; causal sufficiency holds in both). QMCBO differs from MCBO only in the model’s graph view and the lifted targets.

**Invariance (Table 8, App. F).** Under an intra-

Table 2: QMCBO vs MCBO (its own harness and protocol; 5 seeds,  $T=100$ ). Final  $Y$  (all tasks maximized) and wall-clock per unit.

| Env                      | Method | Final $Y$          | s/unit |
|--------------------------|--------|--------------------|--------|
| ToyGraph ( $y^*=2.172$ ) | MCBO   | $1.143 \pm 0.271$  | 1410   |
|                          | QMCBO  | $2.150 \pm 0.013$  | 1570   |
| PSAGraph ( $y^*=-4.13$ ) | MCBO   | $-5.153 \pm 0.000$ | 6830   |
|                          | QMCBO  | $-5.153 \pm 0.000$ | 2266   |

cluster misspecification of the model’s view, QMCBO’s trajectories are identical to their correct-graph runs on 10/10 (env, seed) pairs (Cor. 3 realized), while MCBO’s shift on 8/10, in a direction that is not predictable: on these SCMs the wrong graph sometimes *helped*.

**Soundness matters (Table 2).** On TOYGRAPH, whose cluster contains the only intra-cluster edge, the joint cluster mechanism reaches  $2.150 \pm 0.013$  against the 2.172 optimum — hitting it exactly on three of five seeds — where MCBO never improves on its initial design ( $1.14 \pm 0.27$ ): the within-cluster correlation of Prop. 6 is the missing ingredient. On PSAGRAPH both methods converge to the same plateau ( $-5.15 \pm 0.00$ ), QMCBO in  $3.0\times$  less wall-clock (2266 vs 6830 s per unit).

### 5.6 Each family on its own experiments

We compare each quotient method with its base on the experiments the base method’s own paper ran, in that paper’s own metric, running every base through its authors’ released stack. CBO vs QCBO: the 2020 paper’s Toy, Synthetic, and coral-reef ecosystem examples, best attained  $E[Y \mid \text{do}]$  over trials at unit cost, 10 seeds. DCBO vs QDCBO: its stationary, indepen-

Table 3: Realized price of coarsening per family, each pair on the base method’s own experiment suite (final  $Y$ , mean $\pm$ s.e.). Price is signed so that positive means the quotient did worse in the task’s direction.

| Family        | Experiment        | Base final       | Quotient final   | Price   | Task |
|---------------|-------------------|------------------|------------------|---------|------|
| CBO vs QCBO   | Toy               | $-2.16 \pm 0.00$ | $-2.13 \pm 0.03$ | $+0.03$ | min  |
| CBO vs QCBO   | Synthetic         | $-3.14 \pm 0.19$ | $-1.25 \pm 0.07$ | $+1.89$ | min  |
| CBO vs QCBO   | Coral             | $36.08 \pm 0.02$ | $36.42 \pm 0.01$ | $+0.34$ | min  |
| DCBO vs QDCBO | stat ( $T=3$ )    | $-6.14 \pm 0.05$ | $-6.43 \pm 0.02$ | $-0.29$ | min  |
| DCBO vs QDCBO | ind ( $T=3$ )     | $-3.12 \pm 0.06$ | $-5.57 \pm 0.15$ | $-2.45$ | min  |
| DCBO vs QDCBO | nonstat ( $T=3$ ) | $8.03 \pm 0.90$  | $6.33 \pm 0.29$  | $-1.69$ | min  |
| MCBO vs QMCBO | ToyGraph          | $1.14 \pm 0.27$  | $2.15 \pm 0.01$  | $-1.01$ | max  |
| MCBO vs QMCBO | PSAGraph          | $-5.15 \pm 0.00$ | $-5.15 \pm 0.00$ | $-0.00$ | max  |

dent, and nonstationary DBNs at  $T=3$ , best  $Y$  per time slice, 20 seeds. MCBO vs QMCBO: its ToyGraph and PSAGraph tasks, best attained reward over trials, 5 seeds (the paper’s remaining four problems are function networks, where a partition of function nodes has no causal reading, so we leave them out). Every dataset’s DAG and quotient C-DAG are drawn in App. D. Figure 5 overlays the trajectories and Table 3 reports the finals.

The quotient matches or beats its base on six of the eight experiments, including QDCBO on all three of DCBO’s own setups and QMCBO essentially reaching ToyGraph’s optimum ( $2.15 \pm 0.01$  vs  $2.172$ ) where MCBO never improves on its initial design. The two material positive prices are exactly where Prop. 3 predicts them: Synthetic, whose optimum  $\text{do}(B, D)$  splits the domain partition, and, mildly, Coral. On Toy the single coarse arm is not identifiable from the C-DAG — the bow of Fig. 2 — so, per the declared two-tier rule (Eq. 6), QCBO keeps the arm and runs it on the uninformative prior, learning the response from interventional data alone, and still matches CBO ( $-2.13 \pm 0.03$  vs  $-2.16 \pm 0.00$ ).

## 6 Related work

QCBO occupies an unfilled cell in the CBO landscape: it neither assumes the full DAG nor learns one, but takes a coarser, more reliable input and identifies on the quotient (Table 5, App. F). Methods relaxing the full-graph requirement learn structure online at per-trial cost (GACBO maintains a DAG posterior [Mukherjee et al., 2024], CEO hedges a graph posterior through an entropy-based acquisition [Branchini et al., 2023], Durand et al. [2025] identify  $\text{Pa}(Y)$  under a  $Y$ -leaf assumption) or assume a known equivalence class [Park et al.]. Averaging only *dilutes* a wrong edge in proportion to posterior mass, whereas the quotient pays nothing per trial and is exactly invariant to any edit it does not resolve. The identification machinery we rely on is established (C-DAG do-calculus [Anand et al., 2023], C-DMGs over AD-

MGs [Ferreira and Assaad, 2025], cluster-level Markov properties [Wahl et al., 2024], relaxed admissibility [Bytez.com et al., 2025], C-DAGs as discovery background knowledge [Padh and Kilbertus]), but none of it performs optimization. Closest in spirit, Zeitler [2025] coarsen the *outcome* as a cheap proxy [Poloczek et al., 2017, Beckers and Halpern, 2019]. Causal abstraction relates decisions across levels of granularity, and the quotient is such a map, applied to the manipulable *inputs*.

## 7 Discussion and limitations

Theorem 1 is structural: coarsening acts as a hard regularizer on the hypothesis space, trading resolution (Prop. 3) for immunity to a class of misspecifications. Caveats. (i) The invariance belongs to the *quotient*, and the guarantee is about *behavior*: robustness is read off the CBO-to-QCBO contrast. (ii) DCBO and MCBO cannot join the fixed-objective seam (per-environment coupling), so QDCBO and QMCBO are each evaluated within their own family’s harness with the base method’s analytic objective held fixed. (iii) Faithfulness does not transfer to the cluster level [Wahl et al., 2024]. (iv) Unit intervention costs only. (v) Building the exploration set costs one MIS test and one complete-ID check per candidate cluster union, up to  $2^K - 1$  of each on the *quotient* — exponential in clusters, not in variables, and trivial at the cluster counts practitioners assert. (vi) QMCBO assumes causal sufficiency (Prop. 6), as MCBO does. (vii) What self-corrects and what does not: a wrong *prior* can eventually be overcome by interventional data provided the optimal arm remains in the exploration set; a wrong *exploration set* need not self-correct — under a wrong graph, full-DAG CBO’s MIS can permanently exclude the optimal arm (Remark 2), and under a lossy partition QCBO lacks the sub-cluster arm by construction (Prop. 3). Neither failure is repaired by data; refinement repairs the second, and quotient-invisibility prevents the first for intra-cluster errors. A rollback rule bounding the lossless-trigger tax, and discovering the hierarchy from the run’s own data, are natural follow-ups.

## 8 Conclusion

Coarsening runs causal Bayesian optimization on a coarser, more reliable structural commitment with provable invariance to intra-cluster misspecification, at a price that is zero under the conditions of Prop. 4 and, otherwise, correctable by refinement — the fixed partition’s  $\Theta(T)$  regret becomes  $R_T/T \rightarrow 0$  once a visited partition’s exploration set contains an optimal arm (Remark 4). The practitioner dials resolution against

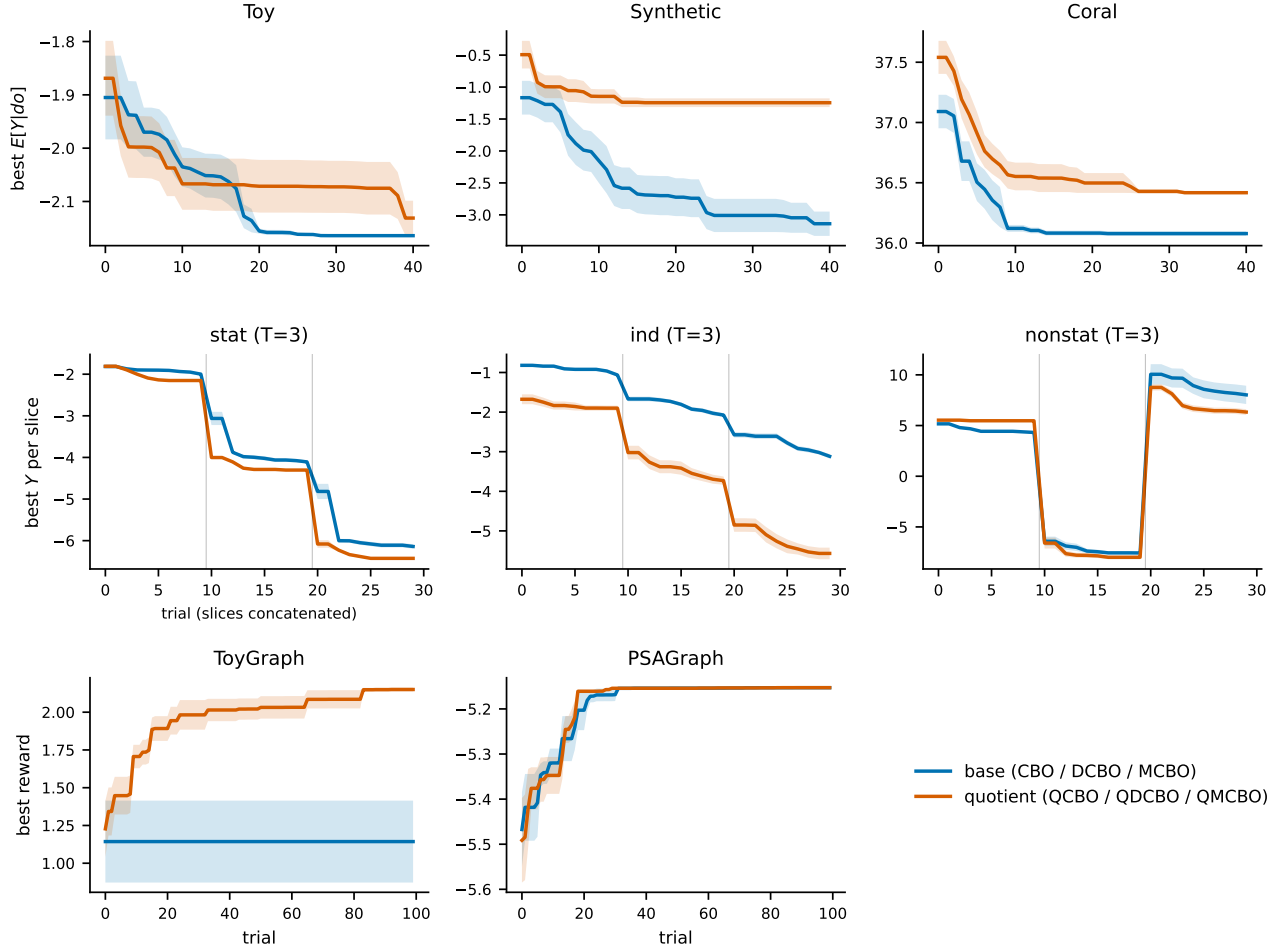


Figure 5: Each quotient method against its base on the base paper’s own experiments and metric (mean $\pm$ s.e.). Top: CBO vs QCBO (best  $E[Y | do]$ , minimization). Middle: DCBO vs QDCBO (best  $Y$  per time slice, trials concatenated across slices). Bottom: MCBO vs QMCBO (best reward, maximization). Dashed line: oracle optimum where known.

robustness by choosing how much structure to commit to, in any of the three families.

## Acknowledgements

## A Proofs

Notation is as in §4.1 and §3.

*Proof of Theorem 1.* First,  $(\mathcal{G}')^\Pi = \mathcal{G}^\Pi$ : the latent projection removes only latent vertices, and an edge between two observables is never an interior vertex of a projection path, so  $(\mathcal{G}')^*$  and  $\mathcal{G}^*$  differ exactly by the modified intra-cluster edges. The quotient keeps a directed (resp. bidirected) cluster-level edge iff some fine edge crosses two distinct clusters, so edits confined within clusters, or redundant across them, leave it unchanged. It remains to show trajectory equality. Write  $\mathcal{A}$ 's run as a sequence of decisions  $d_1, d_2, \dots$  (which arm/action to evaluate, model refits, stopping). By hypothesis each  $d_t$  is a deterministic function  $d_t = f_t((\Pi, \mathcal{G}^\Pi), \mathcal{D}_{t-1}, \omega_t)$  of the quotient, the data collected so far, and the RNG stream. Under  $\mathcal{G}$  and  $\mathcal{G}'$  the first arguments coincide. The objective (the true SCM) and the seeds coincide by the fixed-objective protocol, so  $\mathcal{D}_0, \omega_1$  coincide, hence  $d_1$  coincides, hence  $\mathcal{D}_1$  coincides. Induction over  $t$  gives equality of the entire trajectory.  $\square$

*Proof of Prop. 6.* Under causal sufficiency the fine SCM is Markov relative to  $\mathcal{G}$  with mutually independent exogenous noises, so for any intervention  $\text{do}(\mathbf{x}_S)$  on a union of clusters  $S$  the fine truncated factorization gives  $P(\mathbf{V} \mid \text{do}(\mathbf{x}_S)) = \prod_{k \notin S} P(X_k \mid \mathbf{X}_{\text{pa}_{\mathcal{G}}(k)})|_{\mathbf{x}_S = \mathbf{x}_S}$ . Group the factors by the clusters of  $\Pi$ . For a cluster  $C \not\subseteq S$ , the product of its members' factors is, by the chain rule applied in any topological order of  $\mathcal{G}$  restricted to  $C$ , exactly the joint conditional  $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$ , where  $\text{pa}(C)$ , the union of  $C$ 's parent clusters, contains every out-of-cluster variable appearing in a member's conditioning set (validity of  $\Pi$  guarantees the quotient is acyclic, so  $\text{pa}(C)$  is well defined and precedes  $C$  in C-DAG order). The fine factorization therefore equals the product over clusters of the joint cluster mechanisms with the clusters of  $S$  clamped at  $\mathbf{x}_S$ : the cluster-level truncated factorization. Note both uses of sufficiency: independent noises across clusters (no bidirected edges) and the absence of unobserved common causes that would make a member's conditional depend on variables outside  $\text{pa}(C) \cup C$ . Replacing  $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$  by the product of its per-member marginals would be exact iff the members are conditionally independent given  $\text{pa}(C)$ , in particular when  $C$  contains no intra-cluster edges — hence the joint mechanism in the construction.  $\square$

*Proof of Prop. 7.* Take  $\mathbf{X} = \{X_2, X_3\}$  as one root cluster  $C$ , target  $Y = X_2 + \varepsilon_Y$  its child, all noises standard normal and independent (sufficiency holds, and any  $\Pi$  clustering  $\{X_2, X_3\}$  is valid). Let  $M_1$ :  $X_2 = \varepsilon_2$ ,  $X_3 = aX_2 + \varepsilon_3$ ; and  $M_2$ :  $X_3 = \varepsilon'_3$ ,  $X_2 = a'X_3 + \varepsilon'_2$ , with  $a' = a/(1+a^2)$ ,  $\text{Var}(\varepsilon'_3) = 1+a^2$ , and  $\text{Var}(\varepsilon'_2) = 1-a^2/(1+a^2)$ . Both induce the same zero-mean Gaussian joint on  $(X_2, X_3)$ , with covariance  $\begin{pmatrix} 1 & a \\ a & 1+a^2 \end{pmatrix}$ , hence the same joint cluster mechanism (here, the same root-cluster distribution), and the same  $P(Y \mid \text{do}(\mathbf{x}_C))$  for the whole-cluster intervention (both set  $X_2$  directly). But under the sub-cluster intervention  $\text{do}(X_3 = x)$ : in  $M_1$ ,  $X_2 = \varepsilon_2$  is unaffected, so  $\mathbb{E}[Y \mid \text{do}(X_3=x)] = 0$  for all  $x$ ; in  $M_2$ ,  $X_2 = a'x$ , so  $\mathbb{E}[Y \mid \text{do}(X_3=x)] = a'x \neq 0$  for  $x \neq 0$ ,  $a \neq 0$ . Every quotient-respecting input (the C-DAG, the joint cluster mechanisms, all whole-cluster interventional distributions) coincides between  $M_1$  and  $M_2$ , yet the sub-cluster query differs. No function of those inputs can answer it.  $\square$

*Proof of Prop. 5.* Singleton clusters make the quotient map the identity on  $\mathcal{G}^*$ , so  $\mathcal{G}^\Pi = \mathcal{G}^*$ , unions of singleton clusters are exactly the subsets of  $\mathbf{X}$ , and the MIS condition of Eq. (5) coincides verbatim with Eq. (2). The prior construction poses the same do-calculus queries CBO poses on the full graph, so priors coincide arm-for-arm.  $\square$

**Lemma 1** (MIS reduction). *Let  $G$  be an ADMG with manipulable vertices  $M$  and target  $Y$ , let  $\emptyset \neq \mathbf{W} \subseteq M$ , and set  $\mathbf{W}' = \mathbf{W} \cap \text{An}(Y)$  computed in  $G_{\mathbf{W}}$ . Then (i)  $P(Y \mid \text{do}(\mathbf{W} = \mathbf{w})) = P(Y \mid \text{do}(\mathbf{W}' = \mathbf{w}|_{\mathbf{W}'}))$  for every  $\mathbf{w}$ ; (ii)  $\mathbf{W}'$  is a MIS (or empty, when no member reaches  $Y$ ); and (iii) consequently  $\min_{\mathbf{x}_s \in \text{MIS}(G)} \mu^*(\mathbf{x}_s) = \min_{\emptyset \neq \mathbf{W} \subseteq M} \mu^*(\mathbf{W})$ : restricting the search to MIS loses no value.*

*Proof.* (i) In  $G_{\mathbf{W}}$  the members of  $\mathbf{W} \setminus \mathbf{W}'$  have no directed path to  $Y$ , so  $Y$  is unaffected by their values under the intervention; dropping them from the intervention changes no mechanism on any directed path into  $Y$  (do-calculus rule 3). (ii) A directed path from  $w' \in \mathbf{W}'$  to  $Y$  in  $G_{\mathbf{W}}$  passes through no intervened vertex (their incoming edges are cut), hence through no member of  $\mathbf{W} \supseteq \mathbf{W}'$ ; the same path exists in  $G_{\mathbf{W}'}$ , which only restores edges, so  $\mathbf{W}' \subseteq \text{An}(Y)$  there. (iii) By (i) and (ii) every candidate's value is attained by a MIS, and  $\text{MIS}(G)$  is a subset of the candidates.  $\square$

*Proof of Prop. 1.* Take  $\mathbf{X} = \{a_1, a_2, b_1, b_2\}$  with fine edges  $a_1 \rightarrow b_1$  and  $b_2 \rightarrow a_2$ . The coarsest partition  $\Pi = \{\{a_1, a_2, b_1, b_2\}\}$  is valid: both edges are intra-cluster, so the quotient has a single manipulable node and is acyclic. Its refinement  $\Pi' = \{\{a_1, a_2\}, \{b_1, b_2\}\}$  exposes both edges as inter-cluster ( $a_1 \rightarrow b_1$  contributes  $A \rightarrow B$  and  $b_2 \rightarrow a_2$  contributes  $B \rightarrow A$ ), inducing the 2-cycle  $A \rightleftarrows B$ , so  $\mathcal{G}^{\Pi'}$  is cyclic and  $\Pi'$  is invalid. Decidability is immediate from the acyclicity clause of Def. 1.  $\square$

*Proof of Prop. 2.* (i) Membership at  $\Pi'$  is the stated hypothesis (any union of  $\Pi$ -clusters is a union of  $\Pi'$ -clusters, so the candidate exists at  $\Pi'$ ; being a MIS of  $\mathcal{G}^{\Pi'}$  puts it in  $\mathbf{ES}^{\Pi'}$ ). For the identification functional: quotienting is compositional —  $\mathcal{G}^{\Pi}$  is the quotient of  $\mathcal{G}^{\Pi'}$  under the induced partition of  $\Pi'$ -clusters — so every fine graph compatible with  $\mathcal{G}^{\Pi'}$  is compatible with  $\mathcal{G}^{\Pi}$ . Identifiability from  $\mathcal{G}^{\Pi}$  means a single functional of the observational distribution equals  $\mathbb{E}[Y \mid \text{do}(\mathbf{X}_s)]$  across all SCMs whose graph is compatible with  $\mathcal{G}^{\Pi}$  [Anand et al., 2023]. A fortiori the same functional works across the smaller class compatible with  $\mathcal{G}^{\Pi'}$ . (ii) The ToyGraph counterexample is stated in the proposition; nothing further is required.  $\square$

*Proof of Prop. 3.* By Lemma 1(iii) applied to  $\mathcal{G}^{\Pi}$ ,  $V(\Pi) = \min_{\mathbf{X}_s \in \mathbf{ES}^{\Pi}} \mu^*(\mathbf{X}_s)$  equals the minimum of  $\mu^*$  over *all* non-empty unions of  $\Pi$ -clusters (the value  $\mu^*$  of a cluster union is a property of the underlying SCM, the same at every partition that can express the union). Unions of  $\Pi$ -clusters are unions of  $\Pi'$ -clusters and, at  $\Pi_{\perp}$ , all non-empty subsets of  $\mathbf{X}$ , so the three minima are over nested candidate families:  $V(\Pi_{\perp}) \leq V(\Pi') \leq V(\Pi)$ . Equality  $V(\Pi) = V(\Pi_{\perp})$  holds iff the unrestricted minimum is attained by a union of  $\Pi$ -clusters, and by Lemma 1(iii) at  $\Pi_{\perp}$  the unrestricted minimum is min over Eq. (1)’s exploration set — exactly the stated condition.  $\square$

*Proof of Prop. 4.* We work in the SCM with exogenous variables  $\mathbf{U}$  represented so that all elements of  $\mathbf{U}$  are mutually independent and every latent common cause is a single shared exogenous variable (a bidirected edge in  $\mathcal{G}^*$ ); this is without loss of generality for semi-Markovian models.

Fix a cluster  $C$ , a strict subset  $\mathbf{X}_A \subsetneq C$ , a value  $\mathbf{a}$ , and write  $\mathbf{B} = C \setminus \mathbf{X}_A$ . Let  $\mathbf{B}^{\text{nat}}(\mathbf{u})$  be the value  $\mathbf{B}$  takes under  $\text{do}(\mathbf{X}_A = \mathbf{a})$  at exogenous realization  $\mathbf{u}$ . By the consistency of interventions, setting  $\mathbf{B}$  to the value it would have taken changes nothing realization by realization, so  $Y_{\mathbf{u}}(\text{do}(\mathbf{a})) = Y_{\mathbf{u}}(\text{do}(\mathbf{a}, \mathbf{B} = \mathbf{B}^{\text{nat}}(\mathbf{u})))$ .

*The independence step, made explicit.* Under  $\text{do}(\mathbf{X}_A = \mathbf{a})$ ,  $\mathbf{B}^{\text{nat}}$  is a measurable function of  $\mathbf{U}_{\mathbf{B}}$ , the exogenous variables entering the mechanisms of  $\mathbf{B}$  and of its non-intervened ancestors. At fixed  $(\mathbf{a}, \mathbf{b})$ ,  $Y_{\mathbf{u}}(\text{do}(\mathbf{a}, \mathbf{b}))$  is a measurable function of  $\mathbf{U}_{\mathbf{Y}}$ , the exogenous variables entering the mechanisms of  $Y$  and of its ancestors outside  $C$  (all of  $C$  is intervened in this world, so no path into  $Y$  passes through  $C$ ). Suppose some  $U^* \in \mathbf{U}_{\mathbf{B}} \cap \mathbf{U}_{\mathbf{Y}}$ . Then  $U^*$  has a directed path into some member of  $\mathbf{B}$  — take  $b^*$  to be the *first* cluster member on that path, so the segment  $U^* \rightarrow \dots \rightarrow b^*$  meets  $C$  only at its endpoint — and a directed path into  $Y$  avoiding  $C$  entirely. Concatenating them tail-to-tail yields  $b^* \leftarrow \dots \leftarrow U^* \rightarrow \dots \rightarrow Y$ : a backdoor path from the cluster member  $b^*$  to  $Y$  whose interior avoids  $C$ , i.e. a backdoor path escaping the cluster, which premise (ii) excludes. Hence  $\mathbf{U}_{\mathbf{B}}$  and  $\mathbf{U}_{\mathbf{Y}}$  are disjoint sets of mutually independent exogenous variables, so  $\mathbf{B}^{\text{nat}}$  is independent of the map  $\mathbf{u} \mapsto Y_{\mathbf{u}}(\text{do}(\mathbf{a}, \mathbf{b}))$  for every fixed  $(\mathbf{a}, \mathbf{b})$ .

Taking expectations therefore factorizes into a mixture,  $\mathbb{E}[Y \mid \text{do}(\mathbf{a})] = \int \mathbb{E}[Y \mid \text{do}(\mathbf{a}, \mathbf{b})] dP(\mathbf{B}^{\text{nat}} = \mathbf{b} \mid \text{do}(\mathbf{a}))$ , and a mixture is bounded below by its best component. Premise (i) makes that component reachable: the whole-cluster arm may set  $\mathbf{b}$  anywhere in the natural support, so  $\mu^*(C) \leq \mu^*(\mathbf{X}_A)$ . Applying the argument per cluster covers arms spanning several clusters, hence  $V(\Pi) = V(\Pi_{\perp})$ , and the minimizing whole-cluster values witness a union-of-clusters optimizer.

The two examples below are not decoration: they show the premises are *tight* — dropping either one admits an SCM with strictly positive price, so no weaker hypothesis of this form can give the conclusion.

*Necessity of (i), realized by CLUSTERBENCH10L.* The policy box restricts hard interventions to  $[-2, 2]$  while the strengthened mechanism  $X_2 = 1.5 X_3$  pushes  $X_2$  to  $\approx 2.6$ , near its bowl center 2.7, when  $X_3 \approx 1.7$  is set. The mediated path reaches values the box forbids, so  $\text{do}(X_1, X_3)$  strictly beats every in-box whole-cluster intervention: the realized price is 0.95. (The mixture argument fails at the reachability step: the best component  $\mathbf{b}$  lies outside the interventional domain.)

*Necessity of (ii).* Let  $C = \{A, B\}$  with latent  $U$ , mechanisms  $B = U$  and  $Y = -BU + (A - 1)^2$ ,  $U$  standard normal. Under  $\text{do}(A=1)$  with  $B$  free,  $\mathbb{E}[Y] = -\mathbb{E}[U^2] = -1$ . Under any whole-cluster  $\text{do}(A=a, B=b)$ ,  $\mathbb{E}[Y] = -b\mathbb{E}[U] + (a - 1)^2 \geq 0$ . Intervening severs the exploitable correlation between  $B$  and  $Y$ : the backdoor path

$B \leftrightarrow Y$  escapes the cluster,  $U \in \mathbf{U}_B \cap \mathbf{U}_Y$ , the independence step fails, and the price is 1 despite unconstrained domains.  $\square$

## B ClusterBench10: specification and oracle

**Variables and partition.** ClusterBench10 has five manipulable variables  $\mathbf{X} = \{X_1, \dots, X_5\}$ , four non-manipulable mediators  $\{M_1, M_2, M_3, M_4\}$ , and the target  $Y$ , plus two latent confounders  $U_{12} \rightarrow \{X_1, X_2\}$  and  $U_{15} \rightarrow \{X_1, X_5\}$  (bidirected  $X_1 \leftrightarrow X_2$  and  $X_1 \leftrightarrow X_5$  in the projected ADMG). The coarse partition is  $\Pi = \{C_1, C_2\} \cup \text{singletons}$  with  $C_1 = \{X_1, X_2, X_3\}$ ,  $C_2 = \{X_4, X_5\}$ . All non-manipulable atoms and  $Y$  are singletons. The finest partition  $\Pi_\perp$  (all singletons) recovers CBO (Prop. 5).

**Structural equations.** Roots  $X_1, X_3, X_4, X_5$  carry only exogenous noise (and their latents). The remaining mechanisms are

$$X_2 = 0.3 X_3, \quad M_1 = X_3, \quad M_2 = X_1 + X_2, \quad M_3 = X_4 + X_5, \quad M_4 = X_5,$$

all linear, and the target is a *quadratic bowl* in its parents,

$$Y = \sum_{p \in \text{Pa}(Y)} a_p (p - c_p)^2, \quad \text{Pa}(Y) = \{X_1, X_2, X_3, M_1, M_2, M_3, M_4\},$$

with curvatures  $a = (3.0, 1.5, 1.0, 0.5, 0.8, 0.3, 0.3)$  and centres  $c = (2, 1, 1, 1, 3, 0, 0)$  over  $(X_1, X_2, X_3, M_1, M_2, M_3, M_4)$ . The quadratic form places the optimum at an *interior* point (not a box vertex), so unstructured BO cannot solve it in a constant number of evaluations. This is what makes the sample-efficiency comparison non-degenerate. In the observational sampler every node receives additive Gaussian noise ( $\sigma = 0.7$ , except  $\sigma_Y = 1$  and  $\sigma_{X_3} = \sigma_{X_4} = \sigma_{X_5} = 1$ ), and the latents enter their children with unit loadings,  $U_{12} \sim \mathcal{N}(0, 1.6^2)$  and  $U_{15} \sim \mathcal{N}(0, 1^2)$ . The  $U_{12}$  scale is what produces  $\text{corr}(X_1, X_2) \approx 0.70$ . The latent confounders enter only the observational sampler and the declared structure: the noise-free objective every method optimizes is the deterministic SEM above, identical across all methods and perturbations (App. C).

**Monte-Carlo oracle (verified at build time).** The generator’s oracle minimizes the noise-free objective over each intervention arm’s box and confirms: (C1)  $\text{do}(X_1)$  is the strongest *singleton* arm (it lowers  $Y$  from the no-intervention value  $\approx 23.7$  to  $\approx 3.6$ , a reduction of 20.0, versus 8.1 for  $\text{do}(X_2)$  and 5.6 for  $\text{do}(X_3)$ ), so losing it at the finest partition is a genuine loss; (C2) the global optimum over all subsets is  $\text{do}(C_1) = \text{do}(X_1, X_2, X_3)$  at  $(X_1, X_2, X_3) = (2, 1, 1)$  with  $Y^* = 0$ , a *union of clusters*, so the coarse partition is lossless (Prop. 3);  $C_2$  is neutral-to-harmful (its  $Y$ -terms are minimized at  $X_4 = X_5 = 0$ ), so a method that intervenes on the full 5-dimensional box wastes two dimensions; (C3) the optimum is interior; and the observational sample has  $\text{corr}(X_1, X_2) = 0.70$  via  $U_{12}$ , making the bow a genuine confounded misspecification. All perturbed graphs below are acyclic. The build script is `scripts/generate_clusterbench10.py` and the structural premises are asserted in `ccbo/tests/test_clusterbench10_structure.py`.

**Perturbation taxonomy.** Each perturbation adds/deletes/reverses one or more edges, classified by whether its endpoints lie within a coarse cluster (*intra*) or across clusters (*inter*), and by whether it is *quotient-invisible* (intra, or inter-but-redundant). The headline contrast is **P1** (add  $X_1 \rightarrow X_2$ , an intra- $C_1$  bow on the  $X_1 \leftrightarrow X_2$  confounded pair, which makes  $\text{do}(X_1)$  non-identifiable at the finest partition) versus **Pic** (add  $X_1 \rightarrow X_5$ , the same operation across clusters, which forms a cluster-level bow  $C_1 \rightarrow C_2$  alongside  $C_1 \leftrightarrow C_2$ , rendering  $\text{do}(C_1)$  non-identifiable at the coarse level). The remaining perturbations are P2 (delete  $X_3 \rightarrow X_2$ ) and P3 (reverse it), both intra- $C_1$  prior-bias controls; P5 (delete the sole inter edge  $X_3 \rightarrow M_1$ ) and P6 (delete the *redundant* inter edge  $X_1 \rightarrow M_2$ , with  $X_2 \rightarrow M_2$  surviving so the quotient is unchanged); P7 (omit the latent confounder  $X_1 \leftrightarrow X_2$  from the assumed graph, the *forgotten confounder*: directed edges and hence all identifiability verdicts are untouched, so this is a pure prior-bias perturbation); and a severity sweep S1–S3 of 1, 2, 3 simultaneous intra- $C_1$  edits. QCBO is provably invariant on every quotient-invisible perturbation (P1, P2, P3, P6, P7, S1–S3) and is *not* protected on Pic and P5.

**ClusterBench10D (the damage companion).** Identical structure, mechanisms, noise, and latents: only  $Y$ ’s centers move to the values  $\text{do}(X_1=2)$  induces in the noise-free SEM,  $c = (2, 0, 0, 0, 2, 0, 0)$  over

Table 4: The same intra bow, opposite sign: on CLUSTERBENCH10D the lost arm *is* the optimum’s cheapest route, and full-DAG CBO’s cumulative regret  $R_T$  rises  $4.6\times$  (10 seeds,  $T=100$ ,  $y^*=0$ ).

| Method                   | Assumed DAG        | Final $Y$<br>$\downarrow, y^*=0$ | $R_T$<br>$\downarrow$ | GAP@100<br>$\uparrow$ |
|--------------------------|--------------------|----------------------------------|-----------------------|-----------------------|
| CBO $\equiv$ QCBO-finest | correct            | $0.001 \pm 0.001$                | $2.4 \pm 0.6$         | $0.724 \pm 0.051$     |
| CBO $\equiv$ QCBO-finest | intra bow $P_1$    | $0.014 \pm 0.005$                | $11.1 \pm 2.1$        | $0.724 \pm 0.057$     |
| <b>QCBO-coarse</b>       | either (identical) | $0.066 \pm 0.023$                | $18.5 \pm 4.3$        | $0.809 \pm 0.041$     |

$(X_1, X_2, X_3, M_1, M_2, M_3, M_4)$ . Consequently  $\text{do}(X_1)$  attains the global optimum  $y^*=0$  (the runner-up singleton is  $\geq 14$ ),  $\text{do}(C_1)$  attains the same value (the coarse partition stays lossless), and  $\text{corr}(X_1, X_2) = 0.70$  is unchanged. Build-time oracle checks D1–D3 assert exactly these three premises (`scripts/generate_clusterbench10D.py`). The runner asserts QCBO’s invariance under the bow before writing results.

## C Fixed-objective protocol: per-method details

The invariant is that every method optimizes the *same* objective, the true SCM’s noise-free `intervention_function`, and only the *structure each method reasons with* is perturbed. We realize this per method as follows.

**QCBO (finest and coarse).** The objective is read from the original graph’s SEM, while the structure is supplied separately through the assumed graph name (`CoarsenedGraph(..., assumed_graph_name=variant)`). Swapping the assumed graph changes the quotient, the exploration set, the identification formulas, and the prior, but never the SEM that generates outcomes.

**CBO (full-DAG identification).** CBO is QCBO’s backend at the identity partition (Prop. 5): it runs the same complete identification algorithm, only on the full DAG rather than a quotient. The injection is the same assumed-graph swap as for QCBO (the perturbed structure reaches the exploration set and the identification formulas that build the prior), while outcomes are always drawn from the true SEM. Its trajectory is identical to QCBO’s at the finest partition on every graph.

*Exploration-set disclosure.* Both methods use the MIS of the assumed graph (Eq. 2), computed by the released code and verified against hand-derived Lee–Bareinboim ground truth per benchmark. The CBO authors’ shipped code deviates from that definition in two places, which we correct for both methods: on ToyGraph it lists the non-minimal arm  $\{X, Z\}$  (mutilating  $Z$  cuts  $X \rightarrow Z$ , so  $\text{do}(X, Z) \equiv \text{do}(Z)$  and the joint arm is redundant), and on CoralGraph it truncates the list at subsets of size three (an ad-hoc cap; the definition yields all 31 non-empty subsets, as every manipulable variable keeps a directed path to  $Y$  through never-intervened non-manipulables under any mutilation).

**BO (non-causal control).** BO optimizes a single joint intervention on all manipulable variables with a zero-mean GP prior (`Causal_prior=False`): it reads no structure at all, so it is invariant to every perturbation by construction (we run it once and report  $\Delta = 0$ ), at the price of the weakest absolute performance.

**Byte-identity verification.** For the backend methods the decoupling holds by construction: outcomes are always evaluated on the true graph’s SEM, and the perturbation enters only through the separately-supplied assumed-graph variant. As an independent guard, `seam_ok` evaluates the objective on every singleton arm at random points under the true (P0) versus each perturbed structure and asserts the outcomes match to machine precision. It passes for all perturbations P1–P6 and S1–S3: feeding a method a wrong DAG changes only its causal reasoning, never the problem it is solving.

## D Dataset graphs and their quotients

For every dataset in the paper, the figures below show the fine DAG with its coarse partition and the quotient C-DAG the method actually consumes — regenerated from the released structure registry by `scripts/emit_dataset_tikz.py`, so the pictures cannot drift from the code. The ToyGraph pair is the bow of Fig. 2; on ClusterBench10, note how the intra-cluster latent  $U_{12}$  disappears in the quotient while the cross-cluster  $U_{15}$  survives as  $C_1 \leftrightarrow C_2$  — the exact scope boundary of Theorem 1.



Figure 6: **ToyGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 3: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. The quotient is a bow ( $C_1 \rightarrow Y$ ,  $C_1 \leftrightarrow Y$ ):  $\text{do}(C_1)$  is not identifiable from the C-DAG, so the arm runs on the uninformative prior tier.

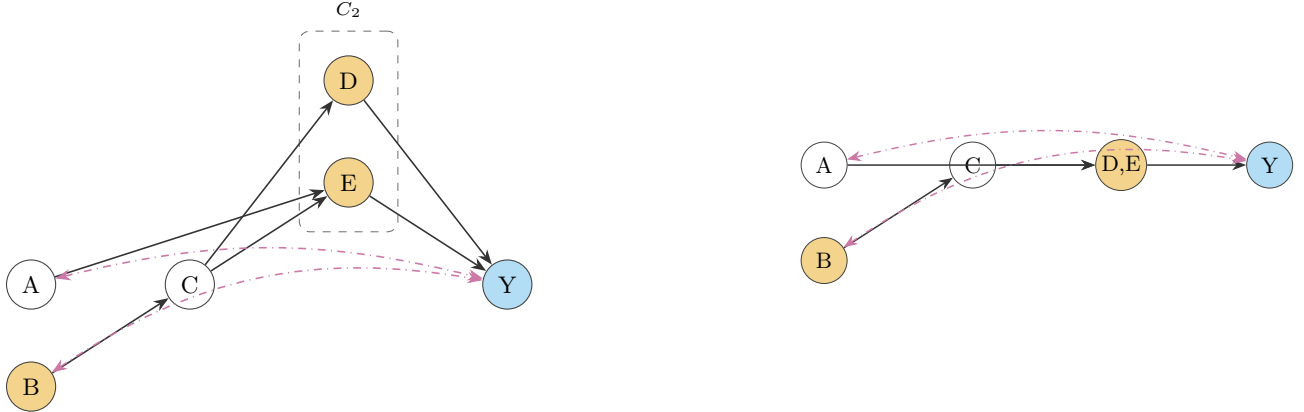


Figure 7: **CompleteGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 3: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

## E Implementation and reproducibility

All methods share one surrogate/acquisition backend (the vendored CBO loop), so comparisons differ only in the assumed structure. **Surrogate.** Each exploration-set arm carries a GPY **GPRegression** model with an RBF kernel (signal variance and lengthscale initialized to 1, ARD off) and observation noise fixed near zero ( $10^{-10}$ , matching the deterministic mechanisms). The causal prior supplies the do-calculus estimate as the GP *mean function*, so  $\Delta$ -metrics isolate the effect of a wrong prior/exploration set from surrogate noise. **Acquisition.** Causal Expected Improvement (Eq. 3), maximized over the box-constrained intervention domain by gradient-based L-BFGS. **Protocol.** Unit per-variable intervention costs, 5 initial interventional points per arm,  $T=100$  trials, 10 seeds. Metrics are defined in §5.1;  $\Delta$ -signs are metric-specific — for GAP a positive  $\Delta$  means the perturbed run converged faster, while for  $Y_T$  under minimization a negative  $\Delta$  means it reached a better value. **Invariance tolerance.** Trajectories are numerically identical on all seeds of every protected perturbation except one seed of the intra bow, where GP-solver floating-point nondeterminism (independent of the perturbation) perturbs intermediate iterates by  $< 10^{-8}$  while the recovered optimum is exact. Every quotient-visible perturbation moves trajectories by  $\geq 10^{-2}$ . We report invariance at this tolerance. **Provenance.** The ClusterBench10 misspecification results in this appendix were produced with the released generators and seeds under an earlier exploration rule in which non-identifiable arms were *removed* from the search set; the released algorithm instead retains them on the uninformative prior tier (Eq. 6). QCBO-coarse’s invariance results are unaffected (the quotient, and hence every verdict, is identical either way); the CBO rows realize the search-set-change channel of Remark 2 as arm removal. The family-suite results (§5.6) use the released self-contained stack. **Artifacts.** The ClusterBench10, ClusterBench10L, and ClusterBench10D generators (with their C1–C3 / L1–L3 / D1–D3 oracle checks), the structure-assertion tests, the `seam_ok` guard, every per-method runner, and the table/figure emitters are released, so all reported numbers regenerate from the released SCMs and seeds.

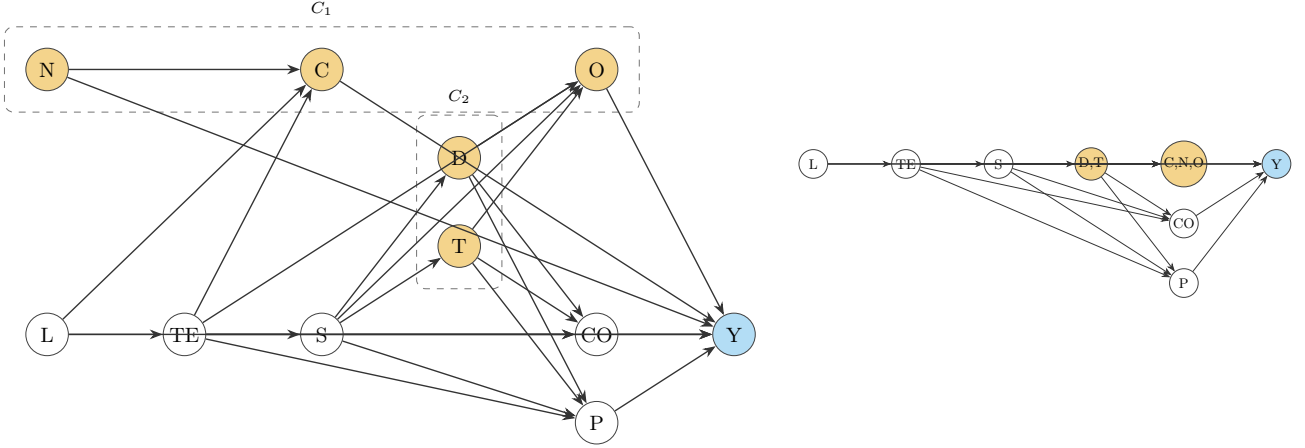


Figure 8: **SimplifiedCoralGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 3: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

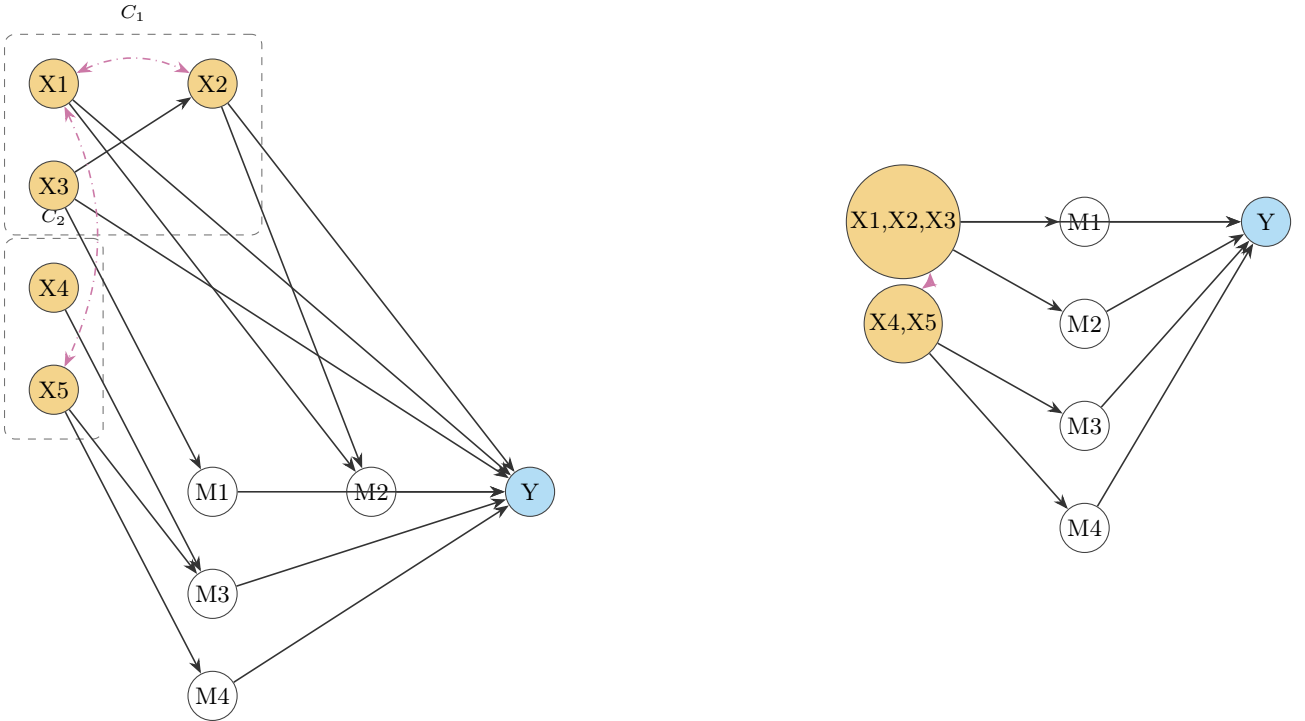


Figure 9: **ClusterBench10 (misspecification suite)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 3: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

## F Full results

Table 9 reports, for every method and every perturbation, the absolute GAP at horizons 20/50/100 and the paired degradation  $\Delta\text{GAP}@100$  and  $\Delta\text{Final } Y$  (misspecified – correct), mean  $\pm$  s.e. over 10 seeds at  $T=100$ . The CBO row is QCBO at the finest partition  $\Pi_{\perp}$  (Prop. 5). Table 10 reports the corresponding regret: the simple regret  $r_T$  of the final incumbent and the cumulative regret  $R_T$  of the incumbent sequence (§5.1).

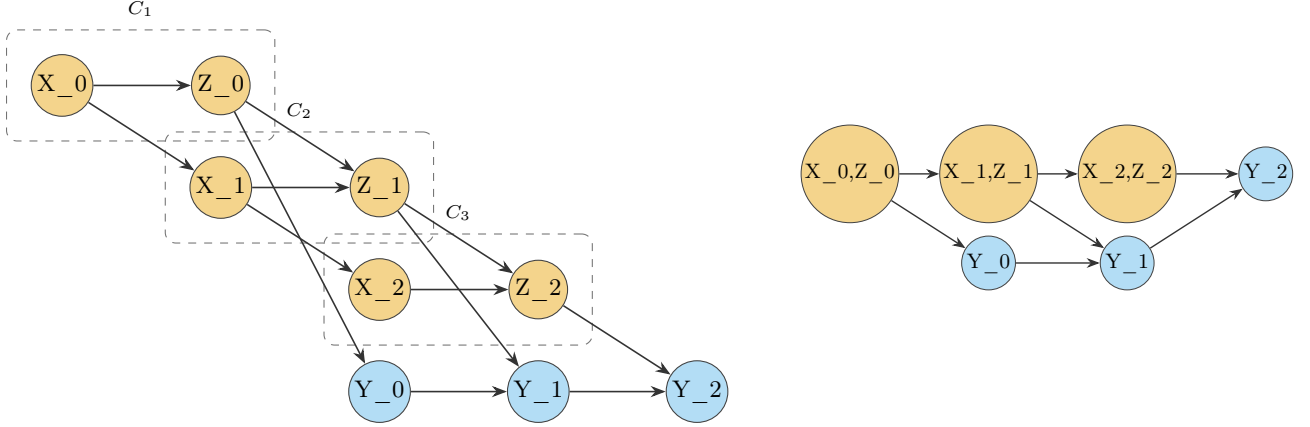


Figure 10: **DCBO stat / nonstat (QDCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 3: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. Slices  $t = 0, 1, 2$ ; the nonstat setup shares this topology with a change point in the SEM.

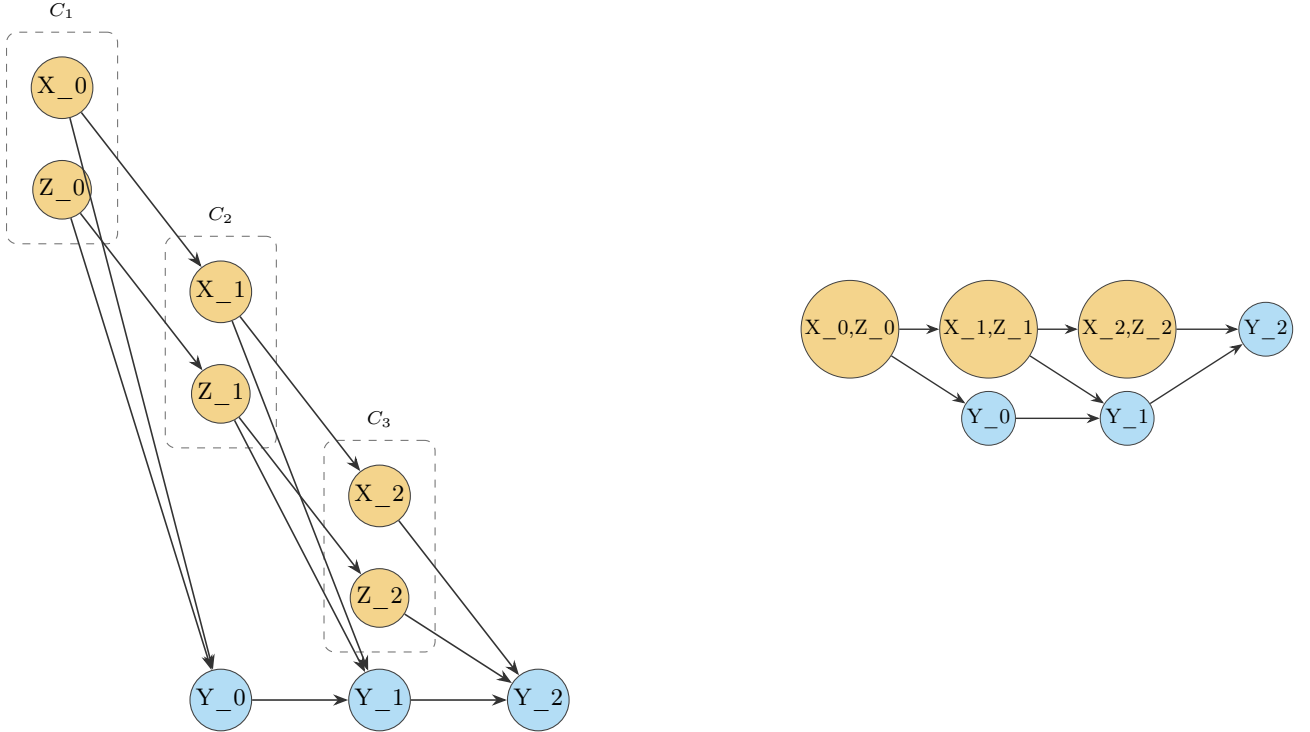


Figure 11: **DCBO ind (QDCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 3: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

## G The cost of a wrong partition

QCBO consumes only the C-DAG and never the fine graph, so a wrong *partition* is stressed by quotienting the base structure once under  $\{X_1, X_2, X_4\}, \{X_3, X_5\}$  ( $X_3$  and  $X_4$  swapped) to obtain a fixed, acyclic but incorrect C-DAG, and optimizing the true objective through it. The wrong partition alters the exploration set, identification verdicts, and priors, but not the objective. By construction it is invariant to the fine-structure perturbations of §5.3, so it has a single cost, reported in Table 11: final quality degrades ( $Y = 0.49$  vs  $0.13$  correct-partition,  $0.29$  CBO) while GAP stays within  $0.06$  of the correct-partition QCBO run ( $0.71$  vs  $0.77$ ), well above CBO ( $0.56$ )



Figure 12: **ToyGraph-M (QMCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 3: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

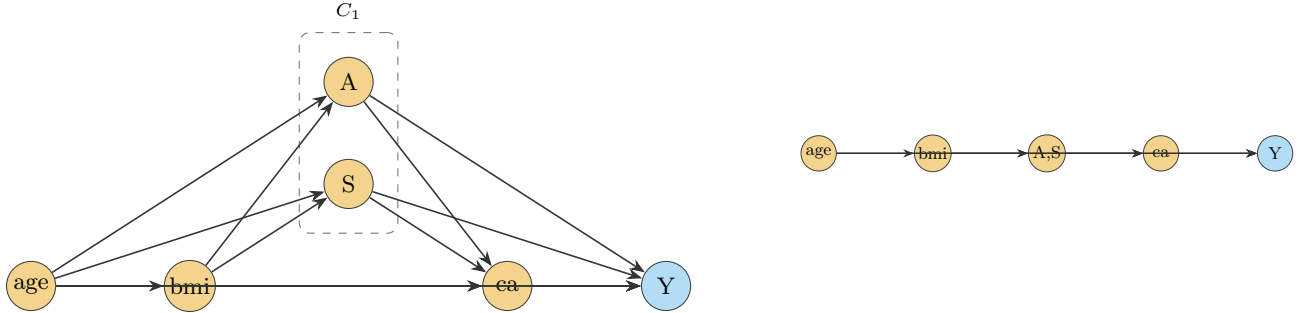


Figure 13: **PSAGraph (QMCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 3: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

and BO (0.18).

## H HQCBO across datasets

Table 12 and Fig. 16 give the full picture for HQCBO (§3.1) on the two CLUSTERBENCH variants (10 seeds each; the finest-partition run supplies the CBO row by Prop. 5). Both regimes are visible. *Lossless* (CLUSTERBENCH10): both quotient methods beat CBO on final  $Y$  and cumulative regret, and the false trigger (HQCBO cannot observe  $y^*$ , so it fires anyway) costs nothing ( $R_T$  41.6 vs QCBO’s 41.8). *Lossy* (CLUSTERBENCH10L): fixed-partition QCBO plateaus with linearly growing regret, while refinement escapes to the CBO-level optimum and bends  $R_T$  back. One honest read follows: refinement lowers GAP relative to fixed-partition QCBO (the mid-run split adds arms and re-explores, trading early-budget GAP for final-value and regret recovery), so it is the right default only when the partition is suspected lossy.

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**Algorithm 1** QCBO: Quotient Causal Bayesian Optimization

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**Require:** C-DAG  $\mathcal{G}^\Pi$  over the clusters of  $\Pi$  (or a fine hypothesis  $\mathcal{G}$  and  $\Pi$ ); target  $Y$ ; number of trials  $T$ ; observational data  $\mathcal{D}^O$ ; optional hierarchy  $H$  (§3.1).

- 1: If given  $\mathcal{G}$ :  $\mathcal{G}^* \leftarrow$  latent projection of  $\mathcal{G}$ ;  $\mathcal{G}^\Pi \leftarrow$  quotient of  $\mathcal{G}^*$  under  $\Pi$ . Check acyclicity of  $\mathcal{G}^\Pi$  (Assumption 1).
- 2: Build  $\mathbf{ES}^\Pi \leftarrow \text{MIS}(\mathcal{G}^\Pi)$  via Eq. (5).
- 3: For each  $\mathbf{X}_s \in \mathbf{ES}^\Pi$ : run complete ID on  $P(Y \mid \text{do}(\mathbf{X}_s))$  and fit the prior mean  $m_s$  via the two-tier rule, Eq. (6), on  $\mathcal{D}^O$ .
- 4: Collect initial interventional data  $\mathcal{D}_0^I$ .
- 5: **for**  $t = 1, \dots, T$  **do**
- 6:   Update GP posteriors  $(m_{s,t}, k_{s,t})$  for all  $\mathbf{X}_s \in \mathbf{ES}^\Pi$ .
- 7:    $(\mathbf{X}_{s_t}, \mathbf{x}_t) \leftarrow \arg \max_{s, \mathbf{x}} \text{EI}_t^s(\mathbf{x}) / \text{Co}(\mathbf{X}_s, \mathbf{x})$ .
- 8:   Intervene, observe  $y_t$ ; update  $\mathcal{D}_t^I$ .
- 9:   **if**  $H$  declared **and** incumbent plateaued: split the incumbent’s clusters via  $H$ , re-check validity, rebuild  $\mathbf{ES}^\Pi \cup$  past arms, carry  $\mathcal{D}_t^I$  by arm, init new arms (§3.1).
- 10: **end for**
- 11: **return**  $(\mathbf{X}_{s_{t^*}}, \mathbf{x}_{t^*})$  where  $t^* = \arg \min_t y_t$ .

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Table 5: Positioning of QCBO against the nearest methods.

| Method                              | What is learned                     | Structural prior  | Search space                |
|-------------------------------------|-------------------------------------|-------------------|-----------------------------|
| CBO [Aglietti et al., 2020]         | nothing (oracle DAG)                | full DAG          | MIS                         |
| GACBO [Mukherjee et al., 2024]      | posterior over DAGs                 | DAG family        | union of MIS                |
| Durand et al. [Durand et al., 2025] | $\text{Pa}(Y) \subseteq \mathbf{X}$ | $Y$ leaf          | $\mathbf{X}$                |
| Park et al. [Park et al.]           | $\text{PAG} \rightarrow \text{MAG}$ | MEC               | partial $\mathbf{X}$        |
| SCB [Lee and Bareinboim, 2018]      | nothing (oracle DAG)                | full DAG          | POMIS                       |
| QCBO (ours)                         | nothing                             | partition + C-DAG | identifiable cluster unions |

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Table 6: Sample efficiency on ClusterBench10 under the *correct* DAG (mean $\pm$ s.e., 10 seeds,  $T=100$ ; lower  $r_T/R_T$  and higher GAP are better).

| Method      | Final $Y$<br>$\downarrow, y^*=0$ | $r_T$<br>$\downarrow$ regret | $R_T$            | GAP@20          | GAP@50<br>$\uparrow$ secondary | GAP@100         |
|-------------|----------------------------------|------------------------------|------------------|-----------------|--------------------------------|-----------------|
| BO          | $1.981 \pm 0.26$                 | $1.98 \pm 0.26$              | $205.6 \pm 27.1$ | $0.05 \pm 0.05$ | $0.15 \pm 0.08$                | $0.17 \pm 0.09$ |
| CBO         | $0.290 \pm 0.05$                 | $0.29 \pm 0.05$              | $71.3 \pm 5.6$   | $0.46 \pm 0.06$ | $0.52 \pm 0.07$                | $0.56 \pm 0.03$ |
| <b>QCBO</b> | $0.128 \pm 0.02$                 | $0.13 \pm 0.02$              | $41.8 \pm 5.2$   | $0.73 \pm 0.05$ | $0.89 \pm 0.02$                | $0.77 \pm 0.07$ |

Table 7: Realized price of coarsening on CLUSTERBENCH10L (lossy; 10 seeds,  $T=100$ ,  $y^*=0.10$ ). The optimum splits  $C_1$ : QCBO plateaus above  $y^*$  by the price of Prop. 3. HQCBO splits  $C_1$  at the trigger and recovers  $y^*$ ; its cumulative regret stops growing after the split.

| Method       | Final $Y$<br>$\downarrow$ | Final $Y - y^*$<br>price $\downarrow$ | $R_T$<br>$\downarrow$ | GAP@100<br>$\uparrow$ |
|--------------|---------------------------|---------------------------------------|-----------------------|-----------------------|
| BO           | $1.188 \pm 0.122$         | $+1.090$                              | $153.3 \pm 32.6$      | $0.765 \pm 0.044$     |
| CBO          | $0.103 \pm 0.001$         | $+0.004$                              | $5.3 \pm 0.7$         | $0.740 \pm 0.045$     |
| <b>QCBO</b>  | $1.128 \pm 0.016$         | $+1.030$                              | $120.3 \pm 3.1$       | $0.802 \pm 0.036$     |
| <b>HQCBO</b> | $0.101 \pm 0.001$         | $+0.002$                              | $32.1 \pm 2.6$        | $0.694 \pm 0.037$     |

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Table 8: Invariance under an intra-cluster misspecification of the model’s graph view (same seeds, deterministic runs): trajectories identical to the correct-graph run, and mean  $|\Delta_{\text{final}}|$ . QMCBO cannot move (Theorem 1). MCBO does.

| Env      | Method | identical | $ \Delta_{\text{final}} $ |
|----------|--------|-----------|---------------------------|
| ToyGraph | MCBO   | 2/5       | 0.299                     |
|          | QMCBO  | 5/5       | 0.000                     |
| PSAGraph | MCBO   | 0/5       | 0.000                     |
|          | QMCBO  | 5/5       | 0.000                     |

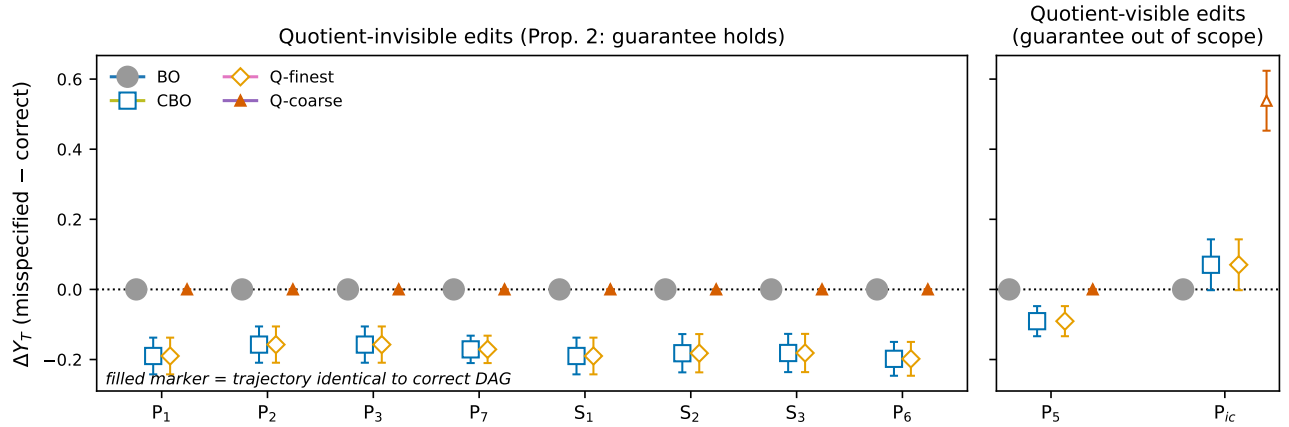


Figure 14:  $\Delta Y_T$  (misspecified - correct) for the whole perturbation field, split at the scope boundary of Theorem 1: quotient-*invisible* edits (left) versus quotient-*visible* (right). Filled markers: trajectories identical to the correct-DAG run. QCBO sits at exactly 0 across the left panel and departs only under the inter bow. The full-DAG method moves on every edge it can see.

Table 9: ClusterBench10: full per-method, per-perturbation results (10 seeds,  $T=100$ ).  $\text{GAP} \in [0, 1]$ , higher is more sample-efficient;  $\Delta$  columns are paired degradations versus the correct DAG, and **ident.** marks identical trajectories.

| Method | Pert. | GAP@20          | GAP@50          | GAP@100         | $\Delta\text{GAP}$ | $\Delta Y_T$       | ident. |
|--------|-------|-----------------|-----------------|-----------------|--------------------|--------------------|--------|
| BO     | P0    | 0.05 $\pm$ 0.05 | 0.15 $\pm$ 0.08 | 0.17 $\pm$ 0.09 | –                  | –                  | ✓      |
|        | P1    | 0.05 $\pm$ 0.05 | 0.15 $\pm$ 0.08 | 0.17 $\pm$ 0.09 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | P2    | 0.05 $\pm$ 0.05 | 0.15 $\pm$ 0.08 | 0.17 $\pm$ 0.09 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | P3    | 0.05 $\pm$ 0.05 | 0.15 $\pm$ 0.08 | 0.17 $\pm$ 0.09 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | P7    | 0.05 $\pm$ 0.05 | 0.15 $\pm$ 0.08 | 0.17 $\pm$ 0.09 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | S1    | 0.05 $\pm$ 0.05 | 0.15 $\pm$ 0.08 | 0.17 $\pm$ 0.09 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | S2    | 0.05 $\pm$ 0.05 | 0.15 $\pm$ 0.08 | 0.17 $\pm$ 0.09 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | S3    | 0.05 $\pm$ 0.05 | 0.15 $\pm$ 0.08 | 0.17 $\pm$ 0.09 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | P6    | 0.05 $\pm$ 0.05 | 0.15 $\pm$ 0.08 | 0.17 $\pm$ 0.09 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | P5    | 0.05 $\pm$ 0.05 | 0.15 $\pm$ 0.08 | 0.17 $\pm$ 0.09 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | Pic   | 0.05 $\pm$ 0.05 | 0.15 $\pm$ 0.08 | 0.17 $\pm$ 0.09 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
| CBO    | P0    | 0.46 $\pm$ 0.06 | 0.52 $\pm$ 0.07 | 0.56 $\pm$ 0.03 | –                  | –                  | ✓      |
|        | P1    | 0.57 $\pm$ 0.05 | 0.63 $\pm$ 0.03 | 0.68 $\pm$ 0.03 | +0.116 $\pm$ 0.043 | –0.190 $\pm$ 0.052 | –      |
|        | P2    | 0.28 $\pm$ 0.07 | 0.58 $\pm$ 0.05 | 0.72 $\pm$ 0.04 | +0.156 $\pm$ 0.039 | –0.157 $\pm$ 0.052 | –      |
|        | P3    | 0.28 $\pm$ 0.07 | 0.58 $\pm$ 0.05 | 0.72 $\pm$ 0.04 | +0.156 $\pm$ 0.039 | –0.157 $\pm$ 0.052 | –      |
|        | P7    | 0.33 $\pm$ 0.06 | 0.57 $\pm$ 0.03 | 0.66 $\pm$ 0.05 | +0.098 $\pm$ 0.038 | –0.171 $\pm$ 0.039 | –      |
|        | S1    | 0.57 $\pm$ 0.05 | 0.63 $\pm$ 0.03 | 0.68 $\pm$ 0.03 | +0.116 $\pm$ 0.043 | –0.190 $\pm$ 0.052 | –      |
|        | S2    | 0.53 $\pm$ 0.05 | 0.58 $\pm$ 0.04 | 0.68 $\pm$ 0.03 | +0.120 $\pm$ 0.049 | –0.182 $\pm$ 0.055 | –      |
|        | S3    | 0.53 $\pm$ 0.05 | 0.58 $\pm$ 0.04 | 0.68 $\pm$ 0.03 | +0.120 $\pm$ 0.049 | –0.181 $\pm$ 0.055 | –      |
|        | P6    | 0.43 $\pm$ 0.04 | 0.62 $\pm$ 0.03 | 0.68 $\pm$ 0.04 | +0.122 $\pm$ 0.052 | –0.198 $\pm$ 0.048 | –      |
|        | P5    | 0.46 $\pm$ 0.05 | 0.56 $\pm$ 0.05 | 0.60 $\pm$ 0.04 | +0.040 $\pm$ 0.032 | –0.091 $\pm$ 0.043 | –      |
|        | Pic   | 0.47 $\pm$ 0.03 | 0.54 $\pm$ 0.04 | 0.57 $\pm$ 0.03 | +0.007 $\pm$ 0.035 | +0.070 $\pm$ 0.073 | –      |
| QCBO   | P0    | 0.73 $\pm$ 0.05 | 0.89 $\pm$ 0.02 | 0.77 $\pm$ 0.07 | –                  | –                  | ✓      |
|        | P1    | 0.73 $\pm$ 0.05 | 0.89 $\pm$ 0.02 | 0.77 $\pm$ 0.07 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | P2    | 0.73 $\pm$ 0.05 | 0.89 $\pm$ 0.02 | 0.77 $\pm$ 0.07 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | P3    | 0.73 $\pm$ 0.05 | 0.89 $\pm$ 0.02 | 0.77 $\pm$ 0.07 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | P7    | 0.73 $\pm$ 0.05 | 0.89 $\pm$ 0.02 | 0.77 $\pm$ 0.07 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | S1    | 0.73 $\pm$ 0.05 | 0.89 $\pm$ 0.02 | 0.77 $\pm$ 0.07 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | S2    | 0.73 $\pm$ 0.05 | 0.89 $\pm$ 0.02 | 0.77 $\pm$ 0.07 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | S3    | 0.73 $\pm$ 0.05 | 0.89 $\pm$ 0.02 | 0.77 $\pm$ 0.07 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | P6    | 0.73 $\pm$ 0.05 | 0.89 $\pm$ 0.02 | 0.77 $\pm$ 0.07 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | P5    | 0.73 $\pm$ 0.05 | 0.89 $\pm$ 0.02 | 0.77 $\pm$ 0.07 | +0.000 $\pm$ 0.000 | +0.000 $\pm$ 0.000 | ✓      |
|        | Pic   | 0.70 $\pm$ 0.04 | 0.73 $\pm$ 0.05 | 0.76 $\pm$ 0.05 | –0.007 $\pm$ 0.084 | +0.538 $\pm$ 0.085 | –      |

Table 10: ClusterBench10: simple regret  $r_T$  ( $\downarrow$ ) and cumulative incumbent regret  $R_T$  ( $\downarrow$ ) per method and perturbation (10 seeds,  $T=100$ ;  $y^*=0$ ).

| Pert. | BO              |                  | CBO             |                 | QCBO            |                |
|-------|-----------------|------------------|-----------------|-----------------|-----------------|----------------|
|       | $r_T$           | $R_T$            | $r_T$           | $R_T$           | $r_T$           | $R_T$          |
| P0    | 1.98 $\pm$ 0.26 | 205.6 $\pm$ 27.1 | 0.29 $\pm$ 0.05 | 71.3 $\pm$ 5.6  | 0.13 $\pm$ 0.02 | 41.8 $\pm$ 5.2 |
| P1    | 1.98 $\pm$ 0.26 | 205.6 $\pm$ 27.1 | 0.10 $\pm$ 0.02 | 65.8 $\pm$ 4.6  | 0.13 $\pm$ 0.02 | 41.8 $\pm$ 5.2 |
| P2    | 1.98 $\pm$ 0.26 | 205.6 $\pm$ 27.1 | 0.13 $\pm$ 0.03 | 62.3 $\pm$ 5.4  | 0.13 $\pm$ 0.02 | 41.8 $\pm$ 5.2 |
| P3    | 1.98 $\pm$ 0.26 | 205.6 $\pm$ 27.1 | 0.13 $\pm$ 0.03 | 62.3 $\pm$ 5.4  | 0.13 $\pm$ 0.02 | 41.8 $\pm$ 5.2 |
| P7    | 1.98 $\pm$ 0.26 | 205.6 $\pm$ 27.1 | 0.12 $\pm$ 0.03 | 58.1 $\pm$ 5.5  | 0.13 $\pm$ 0.02 | 41.8 $\pm$ 5.2 |
| S1    | 1.98 $\pm$ 0.26 | 205.6 $\pm$ 27.1 | 0.10 $\pm$ 0.02 | 65.8 $\pm$ 4.6  | 0.13 $\pm$ 0.02 | 41.8 $\pm$ 5.2 |
| S2    | 1.98 $\pm$ 0.26 | 205.6 $\pm$ 27.1 | 0.11 $\pm$ 0.03 | 98.1 $\pm$ 11.6 | 0.13 $\pm$ 0.02 | 41.8 $\pm$ 5.2 |
| S3    | 1.98 $\pm$ 0.26 | 205.6 $\pm$ 27.1 | 0.11 $\pm$ 0.03 | 98.0 $\pm$ 11.6 | 0.13 $\pm$ 0.02 | 41.8 $\pm$ 5.2 |
| P6    | 1.98 $\pm$ 0.26 | 205.6 $\pm$ 27.1 | 0.09 $\pm$ 0.02 | 43.9 $\pm$ 3.9  | 0.13 $\pm$ 0.02 | 41.8 $\pm$ 5.2 |
| P5    | 1.98 $\pm$ 0.26 | 205.6 $\pm$ 27.1 | 0.20 $\pm$ 0.05 | 59.0 $\pm$ 6.4  | 0.13 $\pm$ 0.02 | 41.8 $\pm$ 5.2 |
| Pic   | 1.98 $\pm$ 0.26 | 205.6 $\pm$ 27.1 | 0.36 $\pm$ 0.08 | 93.2 $\pm$ 4.5  | 0.67 $\pm$ 0.07 | 99.6 $\pm$ 4.0 |

Table 11: Cost of a *wrong* partition on CLUSTERBENCH10 (correct objective; 10 seeds,  $T=100$ ,  $y^*=0$ ). QCBO-wrong $\Pi$  holds a fixed, acyclic but incorrect C-DAG (base structure under a wrong partition). It loses final quality against the correct partition yet degrades gracefully, staying far above non-causal BO.

| Method                          | Final $Y$<br>$\downarrow, y^*=0$ | $r_T$             | GAP@100<br>$\uparrow$ secondary |
|---------------------------------|----------------------------------|-------------------|---------------------------------|
| BO                              | $1.981 \pm 0.261$                | $1.981 \pm 0.261$ | $0.175 \pm 0.090$               |
| CBO                             | $0.290 \pm 0.054$                | $0.290 \pm 0.054$ | $0.561 \pm 0.033$               |
| <b>QCBO</b> (correct $\Pi$ )    | $0.128 \pm 0.021$                | $0.128 \pm 0.021$ | $0.769 \pm 0.066$               |
| QCBO-wrong $\Pi$ (wrong $\Pi$ ) | $0.491 \pm 0.088$                | $0.491 \pm 0.088$ | $0.714 \pm 0.037$               |

Table 12: HQCBO vs BO, CBO, and QCBO across datasets. Final  $Y$  (closest to  $y^*$  best), GAP@100 ( $\uparrow$ ), cumulative incumbent regret  $R_T$  ( $\downarrow$ ); mean $\pm$ s.e., best per group in bold.

| Dataset                                     | Method       | Final $Y$<br>$\rightarrow y^*$      | GAP@100<br>$\uparrow$               | $R_T$<br>$\downarrow$            |
|---------------------------------------------|--------------|-------------------------------------|-------------------------------------|----------------------------------|
| <i>CLUSTERBENCH10</i> (lossless, $y^*=0$ )  |              |                                     |                                     |                                  |
|                                             | BO           | $1.981 \pm 0.261$                   | $0.175 \pm 0.090$                   | $205.6 \pm 27.1$                 |
|                                             | CBO          | $0.290 \pm 0.054$                   | $0.561 \pm 0.033$                   | $71.3 \pm 5.6$                   |
|                                             | QCBO         | $0.128 \pm 0.021$                   | <b><math>0.769 \pm 0.066</math></b> | $41.8 \pm 5.2$                   |
|                                             | <b>HQCBO</b> | <b><math>0.070 \pm 0.021</math></b> | $0.704 \pm 0.059$                   | <b><math>41.6 \pm 5.8</math></b> |
| <i>CLUSTERBENCH10L</i> (lossy, $y^*=0.10$ ) |              |                                     |                                     |                                  |
|                                             | BO           | $1.188 \pm 0.122$                   | $0.765 \pm 0.044$                   | $153.3 \pm 32.6$                 |
|                                             | QCBO         | $1.128 \pm 0.016$                   | <b><math>0.802 \pm 0.036</math></b> | $120.3 \pm 3.1$                  |
|                                             | <b>HQCBO</b> | <b><math>0.101 \pm 0.001</math></b> | $0.694 \pm 0.037$                   | <b><math>32.1 \pm 2.6</math></b> |

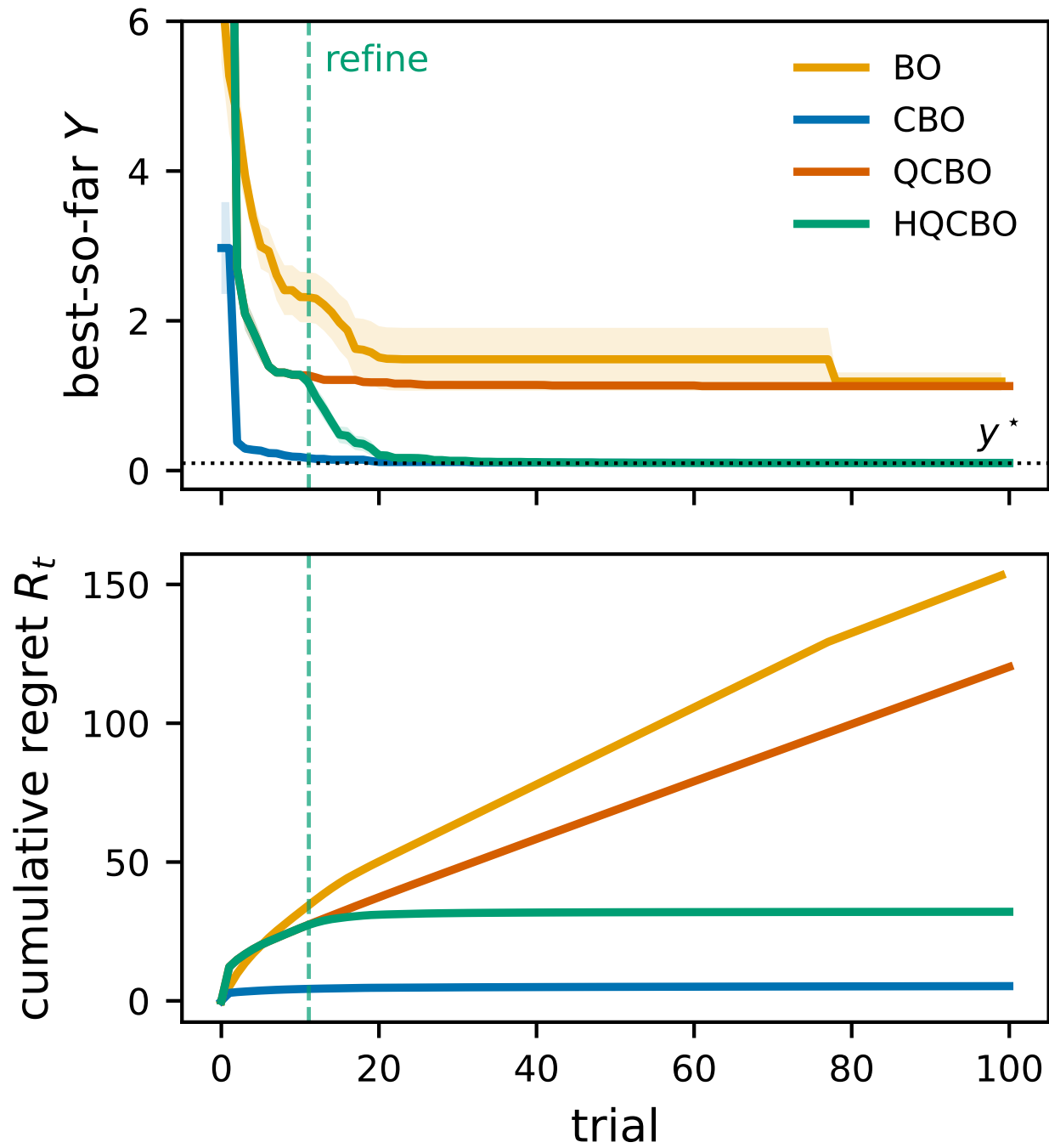


Figure 15: HQCBO on CLUSTERBENCH10L (10 seeds, mean  $\pm$  s.e., dashed vertical line = mean trigger). Top: best-so-far  $Y$ . HQCBO rides the QCBO trajectory, then splits  $C_1$  and descends to  $y^*$ . Bottom: cumulative incumbent regret  $R_t$ . Fixed QCBO's is linear in  $t$ , HQCBO's flattens after the trigger (Remark 4).

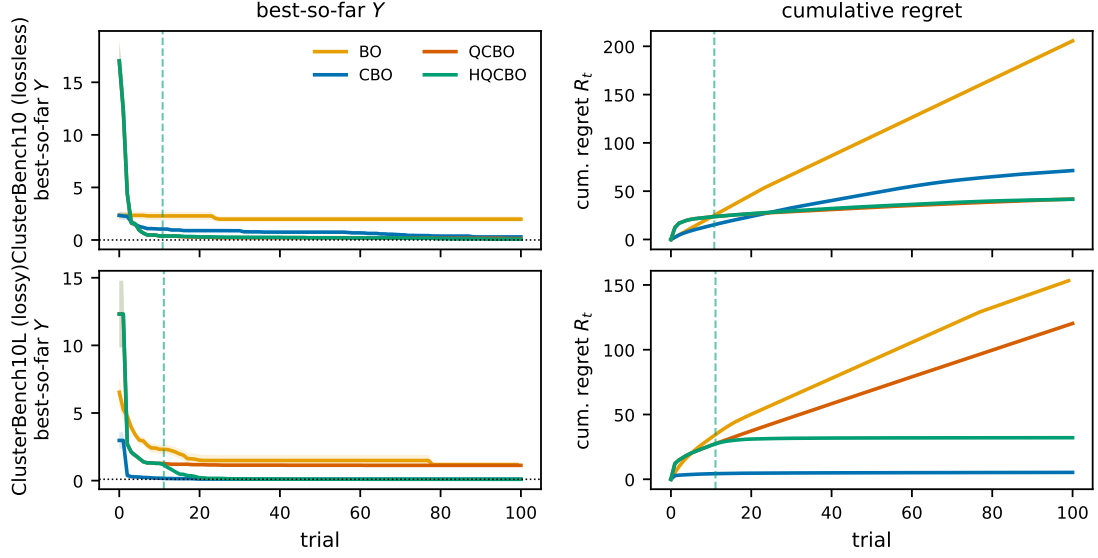


Figure 16: Best-so-far  $Y$  (left) and cumulative incumbent regret  $R_t$  (right) per dataset (mean  $\pm$  s.e., dashed line = mean HQCBO trigger). HQCBO rides QCBO until the trigger, then, where the partition is lossy (CLUSTERBENCH10L), descends and flattens  $R_t$ . Where it is lossless the trajectories stay together.